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THE IMPACT OF REFUGEES ON EDUCATIONAL OUTCOMES OF RESIDENTS

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Abstract

This study links the Canadian waves of the Programme for International Student Assessment (PISA) 2012, 2015, and 2018 ($N \approx 58{,}000$ fifteen-year-olds in 1,379 secondary schools) to a school-level roster of 32 Syrian refugees sampled in 2018 in order to quantify how refugee peers affect resident students' achievement. Refugees are documented to have both negative and positive spillovers through diverse mechanisms. In order to separate and observe these contrasting forces, we implement three identification strategies: (i) a school fixed-effects difference-in-differences design, (ii) a weak-instrument two-stage least-squares variant, and (iii) a doubly-robust Augmented Inverse Probability Weighting (AIPW) estimator that, via cross-fitting, yields conditional average treatment effects (CATEs). We find that the global average treatment effect is small and statistically null once balanced-repeated-replication (BRR) variance is honoured ($\hat{\beta}_{\text{math}} = -0.09$ SD, $\hat{\beta}_{\text{read}} = +0.07$ SD, $\hat{\beta}_{\text{sci}} = +0.04$ SD). However, heterogeneity matters, with CATEs reveal reading gains for low-SES boys in semi-urban schools ($p<0.01$) but mathematics losses for girls in large-city schools ($p<0.01$). Because these subgroup effects rely on school-clustered rather than BRR standard errors, their precision is likely overstated. We conclude that, at Canada's intake intensity (0.15 % of the cohort), Syrian refugees leave the average host student unaffected, yet modest, offsetting effects emerge for specific gender–SES–urbanity combinations. Effective policy should therefore eschew blanket deterrence and instead pair language support for low-income boys with numeracy safeguards for urban girls, while future work pursues larger treated samples or administrative linkages to tighten inference. [1]

[1] I would first like to thank Professor Pena for his guidance and patience, despite the frustrations that I may cause. Additionally, I thank my friends who have inspired me to reach beyond what is readily available, particularly Isha Sharma, Alex Gordon, Ishaan Goel, Redmond Bowcott, and many more who kept me company through many a sleepless night. I finally wanted to thank Professor Ziesler who rekindled my love for math as you see it on full display in my work.

1 Introduction

Images of protests against migrants lined with burning cars and the occasional swastika, headlines linking refugees to crime with undertones of racism, and the steady rise of far-right rhetoric have become familiar fixtures across much of the Western world. In Canada, the United States, and the European Union alike, public debates surrounding refugee policy are often dominated by fear and political calculation. Yet behind these headlines lie quieter, longer-term questions, questions that unfold not in newsrooms or legislatures, but in classrooms among future generations. When governments decide to resettle refugees, they weigh an almost endless list of trade-offs such as economic costs, social cohesion, national values, and the preferences of their constituents. But one crucial consideration often escapes public scrutiny is how do these decisions shape the educational trajectories of the next generation? In a world where technology and education advance in tandem and at light speed, any small effect on the accumulation of human capital when young can have incredible ripple effects, shifting one's utility curve in a permanent way.

This paper focuses on a narrow but important slice of that broader question, how does the presence of refugee students affect their peers in Canadian schools? In particular, I study the case of Syrian refugees admitted during and after the civil war, who were placed across Canada through a mix of government and private sponsorship. While much attention has rightly been paid to the challenges these newcomers face, far less is known about the ripple effects they may create in host classrooms. Do they slow instruction or crowd resources? Do they foster empathy or resilience among their peers? Or are their impacts negligible, overshadowed by other structural inequalities?

Canada's education system offers a rich setting to explore these questions. Syrian arrivals between 2015 and 2018 were large enough to alter the composition of certain schools in meaningful ways, yet dispersed across provinces and school boards with differing capacities and policies. Although private sponsorship introduces quasi-random variation, refugee placement remains correlated with urbanity and immigrant density. To assess their impact, this study links refugee presence at the school or stratum level to student outcomes on the Programme for International Student Assessment (PISA), an internationally benchmarked test administered to 15-year-olds. Because PISA scores track both cognitive skills and future academic prospects through a setting that emphasizes applying skills and logic to real world situations rather than simple memorization, it serves as a valuable measure into the early-life consequences of refugee policy on both the newly arrived and those who welcome them.

Although Canada's private sponsorship stream might appear to generate quasi-random placement—any congregation or family can agree to finance a newcomer for twelve months—we show that refugee assignment is in fact policy-driven and correlated with observable characteristics such as urbanity and pre-existing immigrant density. Simple cross-sectional comparisons would therefore conflate peer effects with selection on school quality. To disentangle the two, we combine three econometric designs: a school fixed-effects difference-in-differences model, a two-stage least-squares variant that treats the DiD indicator as an instrument for immigrant share, and a doubly-robust Augmented Inverse-Probability Weighting (AIPW) estimator that yields both average treatment effects (ATE) and conditional average treatment effects (CATE).

A preview of the findings is instructive. Once the complex PISA survey design is fully respected, the average Canadian student experiences no statistically detectable change in mathematics, reading, or science as a result of Syrian-refugee peers. Beneath that neutrality, however, lies marked heterogeneity: low-income boys in semi-urban schools gain roughly 0.15 standard deviations in reading, while girls in large-city schools lose about 0.20 standard deviations in mathematics. These pockets of benefit and harm are economically modest but educationally salient, and they caution against blanket claims of either crisis or panacea.

The remainder of the paper proceeds as follows. Section 2 sets out a theoretical framework that juxtaposes labour-market incentive and classroom-congestion channels. Section 3 describes the Syrian resettlement architecture and situates it within Canada's decentralised education system. Section 4 introduces the PISA data and constructed covariates. Section 5 details the empirical strategies, emphasising how each design grapples with non-random placement and complex sampling. Section 6 presents the main results, first for the mean effect and then for heterogeneous impacts by gender, socio-economic status, and urbanity. Section 7 discusses policy implications, and Section 8 concludes with limitations and directions for future research.

2 Framework and Model

When governments resettle refugees, they engage in complex utility-maximization. In order to act toward humanitarian obligations, moral commitments, and international norms, they must understand, and balance, the potential economic, political, and social costs of policies enacted. Education is at the heart of this trade-off, as it acts both a medium through which refugee children integrate and a domain in which host communities adjust to newcomers, posing positive and negative impacts between both groups.

Public education systems constantly operate under resource constraints. When schools admit new students, they implicitly reallocate fixed resources such as teacher time, classroom attention, and administrative capacity amongst an evolving population. Refugee arrivals complicate this allocation by both introducing new needs and by modifying peer-group dynamics. In this section, we present a formal framework to explore the mechanisms by which refugee inflows may affect the academic outcomes of resident students.

Let y_{isry} denote the academic performance of student i in school s, located in stratum r, during year y. Each student is assigned to one of two peer groups: treated ($T_r = 1$) or control ($T_r = 0$), depending on whether their school or stratum hosted refugees in 2018. Let A_y be an indicator for the post-treatment period ($A_y = 1$ if $y > 2015$) and R_{ry} be the proportion of first-generation immigrants in stratum r during year y who speak neither french nor english at home.

We conceptualize educational output through the following production function:

$$y_{isry} = f(X_{isry}, R_{ry}, \varepsilon_{isry}), \tag{2.0.1}$$

where X_{isry} includes student and school-level covariates (e.g., parental education, wealth index, urbanization), and ε_{isry} captures unobserved determinants of achievement.

Within the context of our difference in difference model and wald estimator, our parameter of interest is the locally evaluated around the 2016-2018 policy shocks:

$$\tau = \frac{\partial y_{isry}}{\partial R_{ry}}, \tag{2.0.2}$$

in particular for $(\partial R_{ry}|T_r = 1, A_y = 1)$, the change in first-gen immigrant proportion which we attribute to the influx of Syrian refugees. Through this we measure the marginal effect of a change in the immigrant share induced by refugee resettlement on native student outcomes. However, through our AIPW process we attempt to estimate:

$$\tau(X_i) \;=\; \mathbb{E}\big[\, Y_i(1){-}Y_i(0) \;\big|\; X_i \big] \;=\; \mathbb{E}\left[\frac{D_i - \hat{e}(X_i)}{\hat{e}(X_i)\big[1-\hat{e}(X_i)\big]} \big\{ Y_i - \hat{m}_{D_i}(X_i) \big\} + \hat{m}_1(X_i) - \hat{m}_0(X_i) \;\middle|\; X_i \right],$$

where $\hat{e}(X_i)$ is the cross-fitted propensity score and $\hat{m}_d(X_i)$ is the cross-fitted prediction of $\mathbb{E}[Y_i \mid D_i = d, X_i]$ for $d \in \{0, 1\}$. We additionally progress from here to find the conditional average treatment effect to observe the heterogeneity among our population and how different groups are affected differently by exposure to refugees.

2.1 Mechanisms and Trade-Offs

Two principal mechanisms operate in opposing directions. First, labour market incentives suggest that the influx of Syrian refugees led to a substantial increase in the availability of young, low-skilled labour in Western labour markets. This labour supply shock potentially influenced the human capital investment decisions of native adolescents, particularly those who were marginally considering dropping out of school once legal and thus were directly impacted, shifting their decision making over the border. More specifically, this discouragement effect may incentivize boys from lower socioeconomic backgrounds, to remain in school longer and invest more in human capital, thereby raising educational attainment (Tumen 2021).

Second, we observe a classroom congestion effect. Refugee students often arrive with language barriers and trauma histories, placing additional, specialized demands on teachers and classroom management. If schools lack compensatory resources, average instructional quality for existing students may decline (Jensen 2015; Borjas 2007). In addition, they change social dynamics, which can help expand the world view of their peers but also could lead to heightened subdivisions within school dynamics. Within existing literature, immigrants and refugees in large concentrations are associated with negative social dynamic effects such as classroom disruption from language barriers, overcrowding, and cultural frictions that may lower instructional quality or trigger native "flight."

labour market incentives would suggest that an influx of refugees would cause an increase in academic performance from targeted investment in human capital, resulting in a positive τ. Peer-group effects from an influx of refugees, on the other hand, are more ambiguous but are traditionally expected to decrease existing academic performances, indicating a negative τ. Therefore, whether τ is positive, negative, or null depends on the relative strength of these forces and the absorptive capacity of the education system. In high-capacity systems like Canada's, which deploy targeted language support and refugee services, it is theoretically plausible that $\tau \approx 0$, as any potential harm from congestion may be offset by gains through increased effort and civic engagement. Beyond that, we may observe a $\tau > 0$ for certain demographics or the entire population.

3 Institutional Knowledge

3.1 Syrian Civil War and Refugee Crisis

The Syrian civil war emerged from the broader wave of unrest known as the Arab Spring. Long before 2011, Syrians endured strict censorship, pervasive surveillance, and the systematic punishment of political dissent. Severe drought in the late 2000s compounded these pressures, pushing impoverished rural families into cities already simmering with discontent. When security forces tortured teenagers for anti-regime graffiti in Daraa in March of 2011, peaceful demonstrations erupted. Bashar al-Assad's government responded with lethal force, and within months opposition militias had formed, transforming protest into nationwide armed conflict (BBC News 2024).

Escalation was rapid and brutal. International condemnation for Damascus by the United States, the EU, and the Arab League contrasted with diplomatic support from Iran, China, and Russia. By 2021, the war had claimed more than half a million lives, at least two-fifths of them civilian (New York Times 2024). The bloodiest period, 2013 to 2015, accounted for more than 300,000 deaths, or roughly 1.5 percent of Syria's 2011 population of 22 million.

As violence intensified, Syrians fled first to Turkey, Jordan, and Lebanon, and then, in increasing numbers, to Europe and North America. UNHCR data show that Turkey and Jordan had received millions of Syrians by 2012, while Canada began admitting sizeable cohorts only after 2015 (UNHCR 2024). Between January 2015 and June 2018, Canada granted permanent residence to 58,160 Syrians through government-assisted, privately sponsored, and blended programmes (IRCC 2019b).

The demographic profile of these newcomers created immediate educational pressures. According to the 2016 Canadian Census, 51 percent of Syrian refugees were under nineteen (24.4 percent female and 27 percent male) making Syrians one of Canada's youngest refugee populations (Statistics Canada, 2017). Schools therefore faced an urgent mandate to integrate thousands of children rapidly, providing language instruction and cultural orientation. Additionally, due to the historically low immigration of Arabic speakers into Canada compared to other groups, the country heavily struggled with finding well-equipped personnel to help in the integration of refugees.

Globally, the war has displaced more than 14 million Syrians (over half the pre-war population), with 6.8 million now living outside the country (UNHCR 2024). Although the Assad government fell in December 2024, prospects for large-scale returns remain uncertain. Surveys conducted in early 2025 show that, while 80 percent of refugees hope to return eventually, only about 27 percent of those in neighbouring Middle-Eastern states plan to do so in the near term (ReliefWeb 2025). With 70 percent of Syria's remaining residents requiring humanitarian aid and 90 percent living below the poverty line, most Syrians in high-income countries are expected to stay. By analyzing Canada's experience, this paper aims to inform integration policy for current Syrian refugees and any future displacement crises. Canada in particular has an methodical three prong approach that many countries do not adopt and could serve as a model in refugee integration for Asylum seekers if their porcess yields positive educational overflows from refugee integration in schools.

3.2 Canadian Refugee Processing

Canada's response to the Syrian crisis rested on three complementary programs: Government-Assisted Refugees (GARs), Privately Sponsored Refugees (PSRs), and Blended Visa Office-Referred refugees (BVORs). GARs were referred by the United Nations High Commissioner for Refugees (UNHCR 2024) and vetted by Canadian visa officers under criteria that explicitly privileged vulnerability such as survivors of violence or torture, large or female-headed households, and those with pressing medical needs (IRCC 2016). Therefore, GAR processes prioritized households with young children. PSRs, by contrast, entered through the initiative of Canadian citizens and civil-society groups that had raised twelve months of living expenses in advance. While the sponsors selected the applicants (often relatives or acquaintances), each file was still subject to the same security and medical screening as GARs (Labman 2016). As such, we observe a higher concentration of Syrian refugees in immigrant rich areas of Canada such as Ontario, which has 30% of its population as foreign-born immigrants. The BVOR program bridged these two streams, matching UNHCR identified families with Canadian sponsors and splitting first-year financial support between the federal government and the sponsoring group (IRCC 2016).

Across all pathways, the demographic profile of arrivals was strikingly young: nearly half of Syrian newcomers were under the age of fifteen, although unaccompanied or separated minors represented fewer than 4 percent of cases (Statistics Canada 2019). Those few unaccompanied children typically made inland asylum claims and therefore bypassed the mass-resettlement channels, leaving provincial child-protection agencies to improvise guardianship arrangements that critics describe as uneven (Ratkovic et al. 2017).

Because GARs depend on publicly funded services, Immigration, Refugees and Citizenship Canada (IRCC) assigned families to destinations through a "destination-matching process." Officials weighed language capacity, medical requirements, housing costs, and the presence of local settlement agencies before directing newcomers to one of thirty-six Resettlement Assistance Program (RAP) cities. This strategy prevented Toronto and Montréal from absorbing a disproportionate share while ensuring that smaller centres were not overwhelmed (Office of the Auditor General of Canada 2017). As such, the government attempted to spread out refugee allocation, which ultimately was not entirely successful.

The resulting settlement pattern was nonetheless concentrated in Canada's four largest provinces. By November 2016, Ontario had welcomed 39 percent of Syrian refugees; Québec, 24 percent; Alberta, 12 percent; and British Columbia, 11 percent (IRCC 2018). Smaller urban areas, like Halifax, Charlottetown, and Fredericton, successfully lobbied for allocations, hoping to rejuvenate local labour markets and reverse population decline. PSRs and BVORs followed their sponsors instead of federal allocations, producing clusters in Southern Ontario's church networks, prairie towns with long sponsorship traditions, and British Columbia's Lower Mainland (Hyndman et al. 2017). As such, we can assume an exogenous treatment effect through these private sponsors.

The first twelve months after arrival set the tone for long-term integration. GARs received monthly income equal to provincial social assistance rates, along with start-up grants for furniture and clothing. The Resettlement Assistance Program paid approximately $164 million to Syrian households between 2016 and 2018 (IRCC 2019a). Average hotel stays of 19 days preceded moves into private rentals or, less frequently, social-housing units.

PSRs avoided temporary accommodation altogether when sponsors secured housing in advance. In addition to covering living costs, sponsors provided day-to-day mentoring that accelerated tasks such as school registration and job searches (Labman 2016). Similarly, BVOR families benefited from the hybrid arrangement with federal income support for six months, followed by sponsor funding and continuous social assistance (IRCC 2016).

Education quickly became the primary arena of integration for the nearly 30,000 Syrian children who arrived during this period. Provincial ministries mandated immediate enrollment in public schools, where most students entered English- or French-as-a-Second-Language (ESL/FSL) classes. In Ontario, 84 percent of Syrian pupils spent at least one semester in specialized language programs (Guo et al. 2019). Large urban boards hired Arabic-speaking liaisons and interpreters, while rural districts relied on remote interpretation services.

To bridge home and school, the federally funded Settlement Workers in Schools (SWIS) program embedded settlement counsellors in more than 3,000 high-immigrant schools, assisting parents with report-card meetings, referrals to health services, and extracurricular enrollment (IRCC 2022). Nevertheless, access to trauma-informed mental-health care remained fragmented outside major metropolitan areas, leaving many war-affected children on waiting lists (Guo et al. 2019).

Differences among sponsorship categories persisted in the classroom. PSR students typically arrived sooner, missed fewer instructional days, and benefited from sponsors who could advocate proactively with school administrators. GAR students, selected for higher vulnerability, often required additional language and counselling services but faced bureaucratic delays in obtaining them, while BVOR students occupied a middle ground (Hyndman et al. 2017). As such, we expect heterogenous effects among peers who were exposed to PSR refugees versus those exposed to GAR refugees. However, it is unfortunately nigh impossible to distinguish between the refugee classes given our data and capacities, as such we leave it to the reader as inspiration for future study.

Despite its celebrated welcome, Canada's resettlement architecture faces five persistent shortcomings. First, federal and sponsor funding ends after twelve months, yet evidence shows that language acquisition, employment, and mental-health stabilization frequently take much longer (Hyndman et al. 2017). Second, soaring rental prices in cities like Toronto and Vancouver have forced many GAR families into overcrowded or remote housing, undercutting access to transit and services (Office of the Auditor General of Canada 2017). Third, sponsor capacity is uneven, such that while many deliver extraordinary assistance, some underestimate the demands of trauma-affected families (Labman 2016). Fourth, culturally adapted mental-health services remain scarce, leaving complex cases inadequately treated (Guo et al. 2019). Finally, no nationally consistent guardianship framework exists for unaccompanied minors, producing ad-hoc solutions that vary by province (Ratkovic et al. 2017).

Canada's mix of publicly funded and privately mobilized pathways enabled the rapid admission of tens of thousands of Syrians while distributing the financial burden across government and civil society. In addition, the government was able to ascertain their constituents' perspective on the refugee crisis through the PSR programs and shift their actions accordingly. In doing so, they minimized their own costs and partially spread them across the population directly. The GAR program offered predictable material support, whereas private sponsorship harnessed community networks to fast-track social integration. Yet the system's success remains contingent on affordable housing, sustained language instruction, and accessible mental-health care-areas where current provisions lag behind demonstrated need. Addressing these gaps is essential if Canada's humanitarian impulse is to translate into durable inclusion for all refugee categories.

3.3 Literature Review

Canada is widely regarded as a welfare-oriented, education-focused state with a flexible youth labour market, and as such it creates a natural labouratory for studying the consequences of refugee inflows on native students. Between 2015 and 2018, Canada admitted over 30,000 school-age Syrian refugees (IRCC 2019b; Statistics Canada 2017), yet surprisingly little is known about how their presence affects classmates' performance.

As such, evidence from other high-income settings points in opposing directions, underscoring the need for methods that address both selection and measurement challenges inherent in PISA data. Empirical cases studying the effects of labour market mechanisms range from U.S. internal migration shocks (Smith 2012; McHenry 2015; Hunt 2017) to European enlargement episodes (Brunello et al. 2020). In contrast, studies around the educational-experience mechanisms find patterns in U.S. districts (Hoxby 1998; Betts and Lofstrom 2000) and in a cross-state study of immigrant density (Borjas 2007) which consistently show that larger immigrant shares depress test scores when newcomers arrive from disadvantaged backgrounds or lack host-country language skills.

Because refugee inflows typically involve both low-skilled labour supply and acute educational needs, the net effect on host students is theoretically ambiguous. Estimating that net effect requires data granular enough to capture peer exposure and econometric tools capable of addressing selection bias. Only a handful of studies examine refugee peer effects directly.

Tumen (2021) exploits regional variation in Turkey to disentangle a labour-market channel from classroom disruption. By doing so, he was able to isolate a labour market effect where a one-percentage-point rise in refugee share raised PISA scores by about 0.009 SD among lower-performing students. Once refugees started attending Turkish schools in 2018, the earlier gains from labour-market competition were offset by classroom disruption once refugees actually entered schools. Although elegant, that difference-in-differences design relies on stratum-level aggregates, omits school fixed effects, and does not correct for PISA's two-stage sampling or subsampling of students, leaving potential selection and measurement biases unaddressed.

Jensen (2015) documents the flip side in Denmark: a ten-percentage-point increase in immigrant share depresses scores by 4–7 PISA points (0.04–0.07 SD), consistent with overcrowding and language-barrier frictions. But ordinary least squares estimates risk conflating unobserved school quality with peer effects, and do not distinguish refugees from other immigrant groups.

Morales (2022) uses within-school, cross-grade variation in a large U.S. district to show that each additional percentage point of refugee share raises non-refugee math scores by about 0.01 SD, with heterogeneous English-language impacts conditional on baseline achievement. Her clustered-SE results demonstrate that compensatory resources, like federally funded ESL staff and refugee-targeted grants, turn potentially negative peer effects positive, highlighting how targeted supports can reverse negative spillovers. Yet a single-district focus limits external validity.

Building on Tumen (2021), the paper investigates whether labour-market competition operates similarly in Canada's more regulated environment, where minimum-school-leaving ages and apprenticeship tracks blunt direct competition for low-skilled jobs. Adapting Morales (2022), it embeds school-level peer exposure in a national comparison, adding methodological rigour through doubly robust estimation. Finally, it tests Jensen's (2015) concern that classroom disruption can erase labour-driven gains, but does so with Canadian data, a context distinguished by stronger welfare supports and decentralised education governance.

By unifying these strands, the study delivers three contributions: the first causal estimate of Syrian-refugee peer effects on Canadian PISA scores; evidence on the relative strength of labour-market and educational-experience mechanisms in a welfare-state setting; and a methodological template for applying AIPW and Wald estimators to complex, survey-weighted education data. Taken together, the results speak not only to Canadian policymakers but also to broader debates on how high-income countries can reconcile humanitarian commitments with educational equity.

4 Data

Our data consists of the PISA data from 2012, 2015, and 2018 in addition to variables I create through my analysis of the data and certain factors within. Furthermore, I standardize my PISA scores using the PISA theoretical average of 500 and transforming all scores as Z-scores from this theoretical PISA mean.

4.1 PISA Data Description and Construction

PISA employs a two-stage, stratified design that combines efficiency with national representativeness. In the first stage, schools are drawn with probability proportional to size (PPS). If a school S enrolls $N(S)$ fifteen-year-olds, its selection probability is

$$\Pr(S) = \frac{N(S)}{\sum_{S'} N(S')},$$

such that larger schools are more likely to enter the sample, and small schools are not over-represented. In the second stage, PISA selects a simple random sample of up to 35 students per participating school; if fewer than 35 are enrolled, all eligible students are tested. The combination of PPS and prior stratification by region, sector, and size reduces sampling variance and improves the precision of national estimates.

Each student i receives a final survey weight

$$w_i = \frac{1}{p_{\text{school},i}} \frac{1}{p_{\text{student},i}} \times \text{ADJ}_{\text{non-resp}} \times \text{ADJ}_{\text{post-strat}},$$

which multiplies inverse selection probabilities by adjustments for non-response and post-stratification. Balanced-repeated-replication (BRR) with 80 replicates and a Fay factor $\kappa = 0.5$ is used for all variance calculations.

Because each respondent answers only a subset of cognitive items, PISA models achievement with a one-parameter logistic (1PL) item-response model,

$$\Pr(X_i = 1 \mid \theta_s, b_i) = \frac{\exp(\theta_s - b_i)}{1 + \exp(\theta_s - b_i)},$$

where θ_s denotes student proficiency and b_i item difficulty. Item parameters are estimated by marginal maximum likelihood under a latent regression,

$$\theta_s = \mathbf{z}_s' \boldsymbol{\gamma} + \varepsilon_s, \qquad \varepsilon_s \sim \mathcal{N}(0, \sigma^2),$$

with $\mathbf{z}_s$ capturing background characteristics. Ten *plausible values* per discipline are then sampled from the posterior density

$$f(\theta_s \mid \mathbf{X}_s, \mathbf{z}_s) \propto \prod_i \Pr(X_i \mid \theta_s, b_i)\, \phi\big(\theta_s - \mathbf{z}_s' \gamma;\, \sigma^2\big),$$

providing multiple imputations of proficiency (OECD 2021, chap. 9). In our analysis we look at the plausible values of students in mathematics, reading and science.

4.2 Additional Data Using PISA

Through the analysis we determine some key variables we use in our models. The first is refugee status of test takers. It is important to acknowledge that not all refugees in a school take the test and some (or even many) schools' refugees may not be sampled to take the test due to the randomness of the selection process. Now for the Syrian refugees who do take the test, we identify them through their country of birth being Syria, as well as both their parents', a status of first generation immigrant, and an age of immigration of at least 8 (and as such after the beginning of the civil war). In doing so, we identify 32 Syrian refugees in our sample of 22653 in 2018 which corresponds to approximately 0.145 percent of the population. This is in line with Canada's macro data which gives us 50,160 Syrian refugees in 2018, corresponding to approximately 0.135 percent of the total population and is slightly higher accounting for the incredibly young Syrian refugee population. From this classification of refugees, we find the percentage and number of refugees by school and by stratum. We additionally create a variable for urbanity for each stratum on a scale of 0 to 1, as the mean score of each school's urban status in the stratum with 1 as a stratum that is entirely a large city (over 1 million people) and 0.2 as a village (under 3,000 people). We also have the urban score of each school divided by 5 such that it is between 0 and 1. We also create a three-pronged classification of mother's highest education using the International Standard Classification of Education (ISCED) measurement. In doing so we give students with mothers who have completed a tertiary or higher education a 1, those with a high school degree 0.5, and those who have done less education a 0. In addition to this, we have a weighted quartile classification of students by ESCS. Finally, we identify first generation immigrants who speak neither English nor French at home as our immigrants of interest. We quantify them both by density within strata and density within schools allowing for more granular analysis when needed.

4.3 Mean and Variance Estimation

Just to give one a better idea of how PISA's data is structured and how to manipulate it in a more mathematical sense, here is a walk through of how the mean is calculated for each discipline and its variance.

Using the given weights w_i, for plausible values $m = 1, \ldots, 10$:

$$\hat{\mu}_m = \frac{\sum_i pv_i^{(m)} w_i}{\sum_i w_i}.$$

For each plausible value $m = 1, \ldots, 10$ and each replicate $r = 1, \ldots, 80$, the replicate-weighted mean is

$$\hat{\mu}_m^{(r)} = \frac{\sum_i pv_i^{(m)} w_i^{(r)}}{\sum_i w_i^{(r)}}.$$

Within-PV sampling variance is

$$\hat{V}_m = \frac{1}{R(1-\kappa)^2} \sum_{r=1}^{R} \left(\hat{\mu}_m^{(r)} - \hat{\mu}_m\right)^2, \quad R = 80,\ \kappa = 0.5.$$

Rubin's rules yield the final mean and total variance:

$$\bar{\mu} = \frac{1}{M}\sum_{m=1}^{M}\hat{\mu}_m, \qquad T(\bar{\mu}) = \underbrace{\frac{1}{M}\sum_{m=1}^{M}\hat{V}_m}_{\text{within}} + \underbrace{\left(1+\frac{1}{M}\right)\frac{1}{M-1}\sum_{m=1}^{M}(\hat{\mu}_m-\bar{\mu})^2}_{\text{between}},$$

with $M = 10$.

4.4 Descriptive Patterns and Their Impacts

Table 1 reports weighted descriptive statistics for PISA 2018; Table 2 presents analogous results for 2015. Several patterns align with the international literature: males outperform females in mathematics, females outperform males in reading, and science shows small gender gaps. The 2015–2018 comparison also reveals modest score declines for both genders, a trend documented in recent OECD briefs.

Building on Tumen (2021), we expect male outcomes to be buffered by the labour-market mechanism: young men are more likely to perceive direct competition from low-skilled refugee labour, which can raise the payoff to staying in school. For females, peer-interaction effects may dominate. Experimental evidence from China (Han and Li 2009) and administrative data from Italy (Modena, Rettore and Tanzi 2022) show that adolescent girls are especially responsive to the performance and aspirations of close female peers. Consequently, negative (or positive) spillovers from refugee classmates may be more pronounced for girls. All subsequent analyses therefore allow treatment effects to vary by gender.

The combination of plausible-value methodology, BRR replication, and augmented background variables produces nationally representative, design-consistent estimates that can be benchmarked against official OECD reporting. Most importantly, the data architecture supports the doubly robust AIPW and Wald estimators introduced in the empirical chapters, allowing a rigorous assessment of Syrian-refugee effects on Canadian student achievement.

Table 1: Descriptive Statistics by Discipline and Gender (PISA 2018, Standardized Scores)

Discipline	**Group**	**N**	**Mean**	**S.E.**	**SD**	**SD S.E.**
Math	Total	22,653	0.12	0.01	0.92	0.01
	Female (ST004D01T = 1)	11,307	0.10	0.02	0.90	0.01
	Male (ST004D01T = 2)	11,344	0.14	0.01	0.95	0.01
Reading	Total	22,653	0.20	0.00	1.00	0.00
	Female (ST004D01T = 1)	11,307	0.35	0.01	0.96	0.00
	Male (ST004D01T = 2)	11,344	0.06	0.00	1.03	0.00
Science	Total	22,653	0.18	0.01	0.96	0.01
	Female (ST004D01T = 1)	11,307	0.20	0.01	0.93	0.01
	Male (ST004D01T = 2)	11,344	0.16	0.02	0.99	0.01

Note: This table presents basic summary statistics for the PISA sample used in the empirical analysis for 2018.

Table 2: Descriptive Statistics by Discipline and Gender (PISA 2015, Standardized Scores)

Discipline	**Group**	**N**	**Mean**	**S.E.**	**SD**	**SD S.E.**
Math	Total	20,058	0.16	0.01	0.88	0.01
	Female (ST004D01T = 1)	10,022	0.11	0.01	0.85	0.01
	Male (ST004D01T = 2)	10,036	0.20	0.01	0.90	0.01
Reading	Total	20,058	0.27	0.01	0.93	0.01
	Female (ST004D01T = 1)	10,022	0.40	0.01	0.89	0.01
	Male (ST004D01T = 2)	10,036	0.14	0.01	0.94	0.01
Science	Total	20,058	0.28	0.00	0.92	0.00
	Female (ST004D01T = 1)	10,022	0.27	0.01	0.89	0.00
	Male (ST004D01T = 2)	10,036	0.28	0.01	0.96	0.01

Note: This table presents basic summary statistics for the PISA sample used in the empirical analysis for 2015.

5 Empirical Strategy

5.1 Difference-in-Difference Model

5.1.1 Model Validity Verification

We first establish if a DiD model is suitable for our paper. After putting our data in 2012 strata terms to make it comparable over the 3 years, we observe the results in Table 3. We notice that the treated strata have a significantly larger immigrant percentage, with the immigration percentage gap between the treated and control strata growing larger over the years.

Table 3: Summary Statistics by Treatment and Year

Treatment	**2012**	**2015**	**2018**
0 (Control)	Schools: 456 Students: 10,674 % Immigrants: 4.18	Schools: 265 Students: 5,462 % Immigrants: 2.69	Schools: 293 Students: 5,714 % Immigrants: 3.83
1 (Treated)	Schools: 429 Students: 10,870 % Immigrants: 6.28	Schools: 494 Students: 14,596 % Immigrants: 7.70	Schools: 528 Students: 16,939 % Immigrants: 10.19

Table describing each year's treatment and control groups attributes. Treatment groups are chosen by looking at the strata with identified refugees in 2018 and mapping all strata to those in 2012 for consistency since they are not completely similar over the years.

To assess the validity of the parallel trends assumption underlying the difference-in-differences (DiD) design, we estimate discipline-specific regressions of standardized PISA scores on a 2015-year indicator using pre-treatment data from 2012 and 2015. These regressions are conducted separately for the treatment and control strata and include stratum-level fixed effects to account for structural and regional composition changes over time. We have also attached the results when we do not account for the stratum fixed effects in Table 4, where we do not see parallel trends, thereby demonstrating their importance and the heterogeneity between strata.

As shown in Table 5, the estimated pre-treatment trends are small and statistically indistinguishable from zero in mathematics, with reading and science exhibiting somewhat greater divergence in direction and magnitude. To more formally evaluate the parallel trends assumption, we estimate a placebo DiD regression using only the pre-treatment period, where the interaction between the 2015 indicator and the treatment strata serves as a simulated treatment. We observe the results in

Table 6. The placebo effect is not statistically different from zero in any discipline, indicating that observed differences in trends are well within the bounds of sampling variation. These results offer no strong evidence of differential pre-treatment trajectories between treatment and control groups, thereby lending support to the validity of the DiD identification strategy.

5.1.2 Weighted–Least–Squares Regression, BRR–Rubin Inference, and Difference-in-Differences Design

To estimate how Syrian-refugee exposure affects Canadian PISA scores, We employ a two-step procedure endorsed by the OECD. We do a design-based variance estimation within each plausible-value (PV) draw followed by Rubin combination across the ten PVs (OECD 2021, chap. 9–10). This framework is embedded in a difference-in-differences (DiD) specification and is augmented with school-clustered standard errors to guard against intra-school correlation.

For each student i in school s and stratum r in year y, let $y_{isry}^{(m)}$ denote the m-th plausible value of achievement and $\mathbf{X}_{isry}$ the vector of covariates, and small $\mathbf{x}_{isry}$ the vector of regressors in total. We run our regression with the following mathematical model in mind:

$$y_{isry} = \beta_0 + \beta_1(A_{iy} \times T_{ir}) + \boldsymbol{\beta}_2'\mathbf{X}_i + f_s + f_r + f_y + \epsilon_{isry},$$

where $\mathbf{X}_i$ are the student's individual covariates, and f_y, f_r, and f_s the year, stratum, and school fixed effects. We also specify:

$$A_{iy} = \begin{cases} 1 & \text{if year } > 2015 \\ 0 & \text{if year } \leq 2015, \end{cases} \quad \text{and } T_{ir} = \begin{cases} 1 & \text{if } \exists \text{ refugee in stratum in 2018} \\ 0 & \text{if not,} \end{cases}$$

For the school fixed effects, both outcome and regressors are weighted-demeaned within each school:

$$y_{isry}^* = y_{isry} - \frac{\sum_{i\in s} w_i y_{isry}}{\sum_{i\in s} w_i}, \qquad \mathbf{x}_{isry}^* = \mathbf{x}_{isry} - \frac{\sum_{i\in s} w_i \mathbf{x}_{isry}}{\sum_{i\in s} w_i},$$

thereby eliminating any common intercept and focusing estimation on within-school contrasts

(Wooldridge, 2021).

As for each plausible value m, the latent-ability model $y_{isry}^{(m)} = \mathbf{x}'_{isry}\boldsymbol{\beta} + u_{isry}$ is estimated by weighted least squares with the full student weight w_i. The estimator $\hat{\boldsymbol{\beta}}^{(m)}$ minimises the weighted residual sum of squares, treating the weights as inverse selection probabilities.

Sampling variability is assessed by re-estimating the model with each of the eighty BRR replicate weights and forming the Fay-adjusted variance matrix

$$\hat{V}_{\mathrm{BRR}}^{(m)} = \frac{1}{R(1-\kappa)^2}\sum_{r=1}^{R}(\hat{\boldsymbol{\beta}}^{(m,r)} - \hat{\boldsymbol{\beta}}^{(m)})(\hat{\boldsymbol{\beta}}^{(m,r)} - \hat{\boldsymbol{\beta}}^{(m)})',$$

where $R = 80$ and $\kappa = 0.5$ (Fay 1989; Rao and Wu 1992).

Because students within the same school share teachers, facilities, and peer environments, we add a cluster-robust CR0 adjustment

$$\hat{V}_{\mathrm{CL}}^{(m)} = (X'X)^{-1}\Big[\sum_{s} X'_s\hat{\mathbf{u}}_s\hat{\mathbf{u}}'_s X_s\Big](X'X)^{-1}$$

(Liang and Zeger 1986). The within-PV variance becomes $\hat{V}_{\mathrm{BRR}}^{(m)} + \hat{V}_{\mathrm{CL}}^{(m)}$, ensuring robustness to arbitrary within-school error correlation while preserving design consistency.

Finally, we combine all of this using Rubin's rules. Let $\bar{\boldsymbol{\beta}}$ be the mean coefficient vector across PVs, $\bar{W}$ the mean within-PV variance, and B the between-PV covariance. Rubin's rules give the total variance

$$V_{\text{total}} = \bar{W} + \left(1 + \tfrac{1}{10}\right)B$$

(Rubin 1987). Standard errors are the square roots of the diagonal elements, and t-ratios are compared with normal critical values owing to PISA's large effective degrees of freedom (OECD 2021).

Within our framework, our standard errors reflect the sampling variation from PISA's stratified two-stage design and any remaining within-school correlation in unexplained achievement. As such, we have DiD model well in line with PISA's guidelines and account for intra-school correlation.

5.2 Two Stage Least Squares (2SLS)

A central challenge in estimating the causal impact of immigrant exposure on student outcomes is the potential endogeneity of peer composition. Strata with higher immigrant density may differ systematically from others in ways that also affect academic performance, for example, in socioeconomic status, urban infrastructure, or school funding. To address this concern, I employ a two-stage least squares (2SLS) strategy that leverages the currently assumed exogenous variation in immigrant density induced by the arrival of Syrian refugees.

The core identifying assumption is that Canada's mix of centrally and non-centrally managed refugee resettlement policy, introduces as-if random variation in peer composition across strata between 2015 and 2018. This is due to a family or commune anywhere being able to sponsor a refugee in addition to the government's attempt to split refugees across the country but primarily the private sponsorship of refugees. I construct a difference-in-differences (DiD) instrument, equal to one if a stratum received refugees in the post-treatment year (2018), and zero otherwise. This instrument is intended to capture policy-driven changes in immigrant share that are orthogonal to school-level sorting or unobserved student characteristics.

In the first stage of the 2SLS procedure, I regress the stratum-level immigrant density on the DiD treatment indicator and a set of controls, including maternal education, household wealth, urbanity, and survey year. This yields predicted immigrant shares that reflect only the variation attributable to refugee resettlement. This is represented by:

$$\underbrace{imm_{ry}}_{\text{first stage}} = \pi_0 + \pi_1 \,(year_y \cdot region_r) + \pi_2 \, year_y + \pi_3 \, region_r + \boldsymbol{\pi}\mathbf{X}_{is} + v_i.$$

In the second stage, I regress standardized PISA test scores on the predicted immigrant share, using the same covariates. This isolates the component of peer effects on academic outcomes that is plausibly exogenous, addressing the threat of omitted variable bias. Shown by:

$$\underbrace{Y_{isry}}_{\text{second stage}} = \beta_0 + \beta_{2SLS}\,\widehat{imm_{ry}} + \beta_2 \, year_y + \beta_3 \, region_r + \boldsymbol{\beta}\mathbf{X}_{is} + \gamma_i,$$

To accommodate the complex sampling design of PISA, I use weighted least squares with student sampling weights in both stages. Standard errors are computed using a combination of Balanced Repeated Replication (BRR) with Fay's adjustment and school-level cluster-robust covariance estimators. Final uncertainty estimates combine both sources of variance. While results shown below reflect a single plausible value, the complete analysis is repeated across all ten plausible values provided by PISA, with final estimates pooled using Rubin's rules to account for imputation uncertainty.

This approach allows for causal interpretation under standard IV assumptions as: relevance (refugee arrivals shift immigrant share), exclusion (refugee exposure affects outcomes only through changes in peer composition), and monotonicity (refugee placement does not reduce immigrant share in any stratum).

To assess the strength of the instrument, I begin by examining the raw covariance between the DiD indicator and the endogenous regressor (immigrant share), which is 0.81 — indicating a strong unconditional relationship. However, when I estimate a first-stage regression using BRR-weighted least squares, with school fixed effects and a full set of covariates, the DiD coefficient is only 0.0037 (SE = 0.0102, t = 0.36), and the BRR-adjusted F-statistic for the instrument is only 0.12. This result clearly indicates that the instrument is weak in the presence of controls.

Importantly, I find that the inclusion of the treatment stratum (region) dummy in the first-stage regression substantially attenuates the predictive power of the DiD instrument. Despite low raw covariance between DiD and treatment region (0.044), the partial correlation collapses once both are included in a saturated model. This suggests that the treatment stratum status alone accounts for nearly all the policy-induced variation in immigrant share, including that driven by refugee resettlement. This result is intuitive, as while refugee placement is both governmentally and personally managed, it is not randomly assigned across space. Instead, allocation decisions are likely based on regional capacity, service infrastructure, and historical settlement patterns for governments, all factors that correlate strongly with stratum fixed effects. On the flip side, personal sponsors are a self selecting group and often of the same community, or similar community, as Syrians. As such, it is intuitive that strata with higher densities of immigrants will sponsor more refugees, and so the additional refugees impact on immigrant percentage becomes an effect entirely captured by the strata itself .

This interpretation aligns with existing research. For example, Hyndman, Schuurman & Fiedler (2014) show that immigrant settlement in Canada is highly urbanconcentrated, driven by the presence of social networks, labour market opportunities, and pre-existing communities. As a result, even centrally placed refugees tend to be located in strata that already had high immigrant density or capacity, reinforcing the explanatory power of the treatment region dummy in the first stage. For the sake of our methodology we assume that our treatment is exogenous, however given what we see this is highly unlikely. We later implement an AIPW approach to build upon this potentially endogenous treatment.

Therefore, although the DiD instrument is supposedly valid, it suffers from insufficient identifying variation conditional on regional controls. This makes the first-stage regression extremely weak, raising concerns about potential weak-instrument bias in the 2SLS estimates. A natural extension of this work would be to explore alternative or complementary instruments that isolate variation in refugee-induced peer composition while retaining sufficient predictive power.

5.3 Wald Estimator

In addition to the full 2SLS specification, I report a Wald estimator as a transparent summary measure of the causal effect of immigrant exposure on student outcomes. Economically, this approach recovers the local average treatment effect (LATE) of refugee-induced changes in immigrant share, leveraging a binary policy shift (refugee arrival) as the instrument. The Wald estimator is well suited for evaluating discrete interventions, particularly when the endogenous variable is policy-driven and variation arises from treatment assignment rather than endogenous selection. This ratio-based estimator isolates the change in test scores per percentage change in immigrant share induced by refugee placement, abstracting from endogenous mobility. It also provides robustness when the first stage of 2SLS is weak, offering a complementary policy-relevant benchmark to the full specification.

5.3.1 Bootstrap Wald Estimation

To address sampling uncertainty while accommodating the complex PISA design, I implement a bootstrap-based Wald estimator. This method computes the Wald ratio as the quotient of reduced-form and first-stage coefficients, using a balanced resampling design that nests BRR-Fay replicate weighting and school-level cluster correction.

Each bootstrap replicate draws a subsample of $N = 2{,}000$ students while maintaining symmetry across the four DiD cells. Specifically, 200 students are selected without replacement from each cell, and the remaining 1,200 are sampled at random from the residual pool. This structure ensures sufficient variation in the DiD instrument across replicates and avoids oversampling from any corner of the design matrix.

In each replicate, I estimate the reduced-form regression of plausible value $Y_{isry}^{(m)}$ on DiD covariates using BRR-weighted least squares, full student weights, Fay factor $\kappa = 0.5$, and school-clustered standard errors. The resulting coefficient on the interaction term is denoted $\hat{\beta}_Y^{(m)}$. An analogous first-stage regression on the stratum-level immigrant share X_{isry} yields $\hat{\beta}_X^{(m)}$. Replicates for which $|\hat{\beta}_X^{(m)}| < 10^{-3}$ are excluded to mitigate weak-instrument pathologies. For all valid replicates, the Wald estimate is computed as:

$$\hat{\tau}_{b,m}^{*} = \frac{\hat{\beta}_Y^{(m)}}{\hat{\beta}_X^{(m)}}.$$

Because both components are estimated using identical weighting, replicate structure, and cluster corrections, the ratio inherits their design-consistent properties.

The procedure is repeated for $B = 500$ bootstrap iterations. For each plausible value m, the resulting bootstrap distribution $\{\hat{\tau}_{b,m}^{*}\}_{b=1}^{B}$ defines the sampling variability. Its mean is the point estimate, and its standard deviation:

$$\operatorname{se}(\hat{\tau}_m) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\tau}_{b,m}^{*} - \bar{\tau}_m\right)^2}$$

is the bootstrap standard error for plausible value m. Because the BRR-cluster structure is embedded in each replicate, this inner variance accounts for the PISA sampling design, while the outer bootstrap captures uncertainty from the joint estimation of the first and second stages.

After repeating the procedure for all ten plausible values, I combine the resulting estimates using Rubin's rules. Let $\bar{\tau} = \frac{1}{10}\sum_{m=1}^{10} \bar{\tau}_m$ denote the overall mean, $\bar{W}$ the average within-PV variance, and B the between-PV variance. Then the total variance is:

$$V_{\text{total}} = \bar{W} + \left(1 + \tfrac{1}{10}\right) B, \qquad \text{se}(\hat{\tau}) = \sqrt{\text{diag}(V_{\text{total}})}.$$

This procedure ensures valid inference under latent ability imputation, complex survey design, and finite-sample variability in the Wald statistic.

The final coefficient $\hat{\tau}$ measures the causal effect of refugee-induced changes in peer composition on native student outcomes. This estimate is interpretable as causal under three conditions. The first is the DiD treatment is exogenous conditional on fixed effects and trends, the second is refugee exposure affects outcomes only through changes in peer composition (exclusion restriction), and finally the third is the first stage remains sufficiently strong in the bootstrap sample to avoid weak-instrument bias.

5.4 Augmented Inverse Propensity Weighting Model

5.4.1 Justification for AIPW in the PISA Context

Canada's refugee-resettlement rules ensure that exposure to Syrian peers is far from random. Government-assisted refugees (GARs) are directed to Resettlement Assistance Program (RAP) cities chosen for their service capacity, whereas privately sponsored (PSR) and blended-visa (BVOR) arrivals cluster in sponsor-preferred, often highly urban, immigrant-dense neighbourhoods (Diagram 1). Schools that host refugee students therefore differ systematically in labour-market opportunities, language supports and school resources from those that do not. A comparison of PISA scores across treated and control schools would conflate these pre-existing advantages with any true peer effect of refugee presence. Moreover, PISA's school-based sampling design introduces an additional source of measurement error. Not all refugee students present in a school will be selected for testing, meaning some schools classified as untreated may, in fact, host refugee peers. This misclassification reinforces the need for an estimator that can recover causal effects under imperfect treatment observability and non-random assignment.

To recover an unbiased estimate of the effect of refugee peers on student achievement, we employ Augmented Inverse-Probability Weighting (AIPW), which corrects both for non-random assignment and for imperfect treatment measurement in a single procedure. More clearly, one component re-balances the distribution of observable school characteristics (average urban status, immigrant share) across treated and control students—akin to creating a synthetic randomized experiment—while the other fits an outcome surface that adjusts any remaining bias in test scores conditional on those characteristics. Crucially, AIPW is doubly robust, meaning consistency is obtained if either the treatment model or the outcome model is correctly specified, with efficiency achieved if both are (Robins, Rotnitzky and Zhao, 1994; Bang and Robins, 2005).

We verify the non-randomness of placement by estimating a logistic model of stratum treatment using PISA's BRR-Fay replicate weights. Let $T_r = 1$ if any Syrian students appear in stratum r in 2018 and $X_r = (\text{urban\_level}_r,\ \text{immigrant\_share}_r)$. The fitted equation

$$\Pr(T_r = 1 \mid X_r) = \frac{\exp(X_r^\top \hat{\gamma})}{1 + \exp(X_r^\top \hat{\gamma})}$$

yields significance with $\hat{\gamma}_{\text{urban}} = 1.34$ ($t = 5.30$), $\hat{\gamma}_{\text{immigrant}} = 3.2$ ($t = 3.45$), and an Efron pseudo-$R^2 = 0.42$ as seen in table 7. More simply, based on the state of the strata in 2015, we are able to determine, to a certain degree, which of the strata become treated in 2018 using only the 2015 share of immigrants in the stratum and its average urbanity. These results confirm that refugee placement is driven by observable context rather than chance, so ignorability cannot be assumed without adjustment. Furthermore, due to the PSR program, we know that the probability of being treated is non-zero for all students (although it may come close) since there are many unknowns that go into the sponsoring of refugees by individuals or communities.

5.4.2 AIPW Implementation with Student Weights and Plausible Values

We now describe how AIPW is implemented using only the original PISA student sampling weights w_i and the ten plausible values $Y_i^{(m)}$ per student, omitting BRR replication for simplicity which we explain in the next. For each student i, let $D_i \in \{0,1\}$ indicate whether their school hosts any Syrian peers,

$$X_i = \big(IMMIG\_DENSITY_i,\ URBAN\_3_i,\ ESCS\_WQUART_i,\ ST004D01T_i\big),$$

and $Y_i^{(m)}$ the mth plausible value of their test score. We include these four covariates because they enter both the propensity-score model $\hat{e}(X_i) = P(D_i = 1 \mid X_i)$ and the outcome model $\hat{\mu}_d(X_i) = E[Y_i \mid D_i = d, X_i]$ in ways predicted by economic theory, thus satisfying at least one of our conditions together to trigger the doubly robust mechanism, since consistency requires only one of these two models to be correctly specified.

- **Immigrant density** ($IMMIG\_DENSITY_i$)**.** In human-capital externality models, the peer composition in a classroom affects individual learning via resource dilution and teacher effort. Empirical studies show that a one-standard-deviation increase in immigrant share lowers native PISA math scores by about 0.05 SD, consistent with a negative classroom congestion effect (Jensen, 2015). We have already explained how immigration density predicts refugee presence as one leads to the other.

- **Urban status** ($URBAN\_3_i$)**.** Agglomeration economies imply higher returns to schooling in cities through knowledge spillovers and better public-good provision (Raul, 2012) documents that urban students score 0.25 SD above rural peers. In a location-choice model, parents trade off housing price $P(L)$ against expected utility $U = E[Y_i \mid URBAN_i]$, so because urban schools differ systematically in peer composition and achievement, omitting urbanity would bias both the propensity score and the outcome model. This it enters the propensity score, and omitting it induces endogeneity if $URBAN\_3_i$ correlates with both D_i and Y_i .

- **ESCS quartile** ($ESCS\_WQUART_i$)**.** According to the Becker–Mincer framework, parental socio-economic status enters the child's educational production function multiplicatively:

$$Y_i = \alpha_0 + \alpha_1 ESCS\_WQUART_i + \alpha_2 X_i + \varepsilon_i,$$

where each quartile shift correlates with a 10–15-point jump in PISA scores (OECD, 2023). Including $ESCS\_WQUART_i$ controls non-linear SES effects on both treatment and outcome. Furthermore, ESCS is a strong variable to be considered in school choice problems, as such it impacts where a child will be schooled. This in turn affects the probability of being in school with a refugee to a certain degree, giving us our two way effect.

- **Gender** ($ST004D01T_i$)**.** Gender differences in test performance arise from social norms and stereotype threat (González de San Román and de la Rica, 2012). PISA data show boys outperform girls by 0.1 SD in math, and girls outperform boys in reading by 0.15 SD. Formally,

$$\mu_d(X_i) = \cdots + \delta\, ST004D01T_i + \dots,$$

so omitting gender when $Cov(D_i, ST004D01T_i) \neq 0$ biases the AIPW estimator.

By conditioning on these covariates, we render the conditional independence assumption plausible and exploit the double-robustness of AIPW.

We first estimate the propensity score $\hat{e}(X_i) = \Pr(D_i = 1 \mid X_i)$ via a random-forest classifier with 200 trees, trained with five-fold cross-fitting: the sample is partitioned into five folds, and for each fold the forest is grown on the other four, so that each student's $\hat{e}(X_i)$ is always out-of-sample. This guarantees that the same data are never used both to fit and to predict, eliminating overfitting bias and avoiding false positives in treatment prediction (Chernozhukov et al., 2018).

Next we fit two random forest regressions, one on the treated subsample to learn $\hat{m}_1(X_i) \approx \mathbb{E}[Y_i \mid D_i = 1, X_i]$, and one on the control subsample to learn $\hat{m}_0(X_i) \approx \mathbb{E}[Y_i \mid D_i = 0, X_i]$ again using five-fold cross-fitting and weighting by w_i.

For each observation we compute the AIPW influence score which measures how much a particular student, with their particular background and observed outcome, pulls the overall ATE up or down:

$$\psi_i^{(m)} = \frac{D_i - \hat{e}(X_i)}{\hat{e}(X_i)\big(1 - \hat{e}(X_i)\big)}\big(Y_i^{(m)} - \hat{m}_{D_i}(X_i)\big) \;+\; \hat{m}_1(X_i) - \hat{m}_0(X_i).$$

Averaging over students with their sampling weights gives the per-PV estimate

$$\hat{\tau}^{(m)} = \frac{\sum_i w_i \, \psi_i^{(m)}}{\sum_i w_i}.$$

Finally, we pool across the ten plausible values by simple mean $\bar{\tau} = \frac{1}{10}\sum_{m=1}^{10} \hat{\tau}^{(m)}$, which corrects for the imputation uncertainty in PISA's latent-ability scoring.

Because the estimator remains consistent when either the propensity or the outcome model is misspecified, and because cross-fitting restores valid asymptotics, AIPW delivers credible causal estimates of refugee peer effects even under complex, policy-driven assignment and imperfect treatment measurement. However, it is important to note that by not factoring in BRR and resulting standard error adjustments, this version of AIPW will heavily underestimate standard errors and attribute causal significance where there may be none. We thus also run a model that has school clustered standard errors as an in-between with which we run our CATE due to computational limits with BRR.

5.4.3 AIPW Fully Integrated with PISA Methodology

To align the AIPW estimator completely with PISA's complex survey design and get a more realistic causal relationship, we embed its two stages—propensity-score weighting and outcome regression—within PISA's balanced-repeated-replication (BRR) framework and plausible-value imputation (Kurz, 2022). First, for each of the ten plausible values $Y_i^{(m)}$ we compute cross-fitted propensity scores $\hat{e}^{(-k)}(X_i)$ and outcome predictions $\hat{m}_d^{(-k)}(X_i)$ using random forests trained on the four other folds (Chernozhukov et al., 2018). This ensures that no student's score enters the fitting of the models used to predict that same student.

Second, we calculate the fold-wise AIPW influence scores

$$\psi_i^{(m)} = \frac{D_i - \hat{e}^{(-k)}(X_i)}{\hat{e}^{(-k)}(X_i)\big(1 - \hat{e}^{(-k)}(X_i)\big)}\big(Y_i^{(m)} - \hat{m}_{D_i}^{(-k)}(X_i)\big) \;+\; \hat{m}_1^{(-k)}(X_i) - \hat{m}_0^{(-k)}(X_i),$$

and aggregate them with the student sampling weight w_i to obtain a point estimate $\hat{\tau}^{(m)}$ for each plausible value.

Third, to propagate PISA's two-stage stratified design variance into our AIPW estimates, we recompute the entire cross-fitting and influence-score averaging on each of the $R = 80$ BRR replicate-weight vectors $w_i^{(r)}$. Denote by $\hat{\tau}^{(m,r)}$ the AIPW estimate for plausible value m under replicate r. The Fay-adjusted within-PV variance is then

$$\hat{V}_{\text{BRR}}^{(m)} = \frac{1}{R(1-\kappa)^2} \sum_{r=1}^{R} \big(\hat{\tau}^{(m,r)} - \hat{\tau}^{(m)}\big)^2, \quad \kappa = 0.5.$$

Finally, Rubin's combining rules produce the overall AIPW point estimate and its standard error by treating the ten plausible-value estimates $\hat{\tau}^{(1)}, \ldots, \hat{\tau}^{(10)}$ as multiple imputations. Let

$$\bar{\tau} = \frac{1}{10} \sum_{m=1}^{10} \hat{\tau}^{(m)}, \qquad \bar{W} = \frac{1}{10} \sum_{m=1}^{10} \hat{V}_{\text{BRR}}^{(m)}, \qquad B = \frac{1}{9} \sum_{m=1}^{10} \big(\hat{\tau}^{(m)} - \bar{\tau}\big)^2.$$

Then the total variance is $\bar{W} + (1 + 1/10)\, B$, and its square root is reported as the AIPW standard error (Rubin, 1987; OECD, 2021).

Because each student's contribution to $\bar{\tau}$ is based on out-of-sample predictions, and because both survey-design uncertainty and imputation uncertainty are explicitly incorporated using BRR and Rubin's rules respectively, the resulting estimator is simultaneously design-consistent, doubly robust, and semiparametrically efficient. In practical terms, the final AIPW estimate can be interpreted as the average causal effect, in PISA score points, of attending a school with Syrian refugee peers for the national cohort of fifteen-year-olds in Canada, free of bias from both policy-driven placement and sampling artifacts.

5.4.4 Conditional Average Treatment Effect Using AIPW

While the average treatment effect (ATE) tells us how, on average, exposure to Syrian refugee peers shifts PISA scores for the entire Canadian cohort of fifteen-year-olds, we know that this effect may differ across students facing different school contexts or backgrounds. As previously established, our labour market effects and class congestion effects vary by gender and by academic investment since students with lower test scores will be more likely to join the low-skilled job market. To uncover such heterogeneity, we turn to the notion of the Conditional Average Treatment Effect (CATE). For a vector of characteristics X_i, the CATE is defined as

$$\tau(x) \;=\; \mathbb{E}\big[Y_i(1) - Y_i(0) \mid X_i = x\big],$$

the average gain in PISA points from attending a school with refugee peers, for all students whose covariates equal x (thus we prioritize discrete covariates).

Augmented inverse-probability weighting naturally yields an estimate of each individual student's contribution to the treatment effect in the influence function score established as:

$$\psi_i \;=\; \frac{D_i - \hat{e}(X_i)}{\hat{e}(X_i)\big(1 - \hat{e}(X_i)\big)}\big(Y_i - \hat{m}_{D_i}(X_i)\big) \;+\; \hat{m}_1(X_i) - \hat{m}_0(X_i)\,,$$

which satisfies $\mathbb{E}[\psi_i \mid X_i] = \tau(X_i)$. In other words, conditional on X_i, the pseudo-outcome ψ_i has expectation equal to the true individual treatment effect.

To estimate the CATE function $\tau(x)$ we therefore proceed in two steps. First, after computing each student's cross-fitted AIPW score ψ_i, we regress those scores on the chosen covariates X_i. This asks how the estimated peer-effects for student i vary with their school's urban status, same gender, similar mother's education, or socioeconomic status. Formally, we fit a flexible second-stage model

$$\psi_i = g(X_i) + \varepsilon_i,$$

where g might be a low-dimensional linear specification with interactions or a richer machine-learning model (for example, a random forest). The fitted function $\hat{g}(x)$ is our CATE estimator $\hat{\tau}(x)$.

Second, we can summarize heterogeneity by evaluating $\hat{\tau}(x)$ at points of economic interest—say, the difference in effect for urban versus rural schools, or for the bottom versus top quartile of immigrant density. For instance, if $X_i = (\text{URBAN}_i,\ \text{IMMIGRANT\_DENSITY}_i)$, we might report

$$\hat{\tau}\big(\text{urban} = 1,\ \text{immig\_dens} = 0.10\big) \;-\; \hat{\tau}\big(\text{urban} = 0,\ \text{immig\_dens} = 0.10\big),$$

to quantify how the peer effect differs between an urban and a rural school at the same 10 percent immigrant density.

Because AIPW guarantees that ψ_i is an unbiased signal for each student's treatment effect whenever either the propensity model $\hat{e}(X)$ or the outcome regressions $\hat{m}_d(X)$ is correct, the two-step procedure yields consistent CATE estimates even under moderate misspecification. Intuitively, the inverse-weighting step ensures that comparisons across treated and control students are fair on observables, while the outcome-regression step flexibly adjusts for any remaining structure in how test scores respond to those observables. By conditioning on X in the second stage, we learn how the peer effect itself varies with student and school characteristics.

In practice, we implement the second-stage regression using the same random-forest machinery (with cross-fitting) that we used for the first stage, ensuring that CATE estimates inherit the same protection against overfitting and spurious heterogeneity (Chernozhukov et al., 2018). Standard errors for CATE estimates can be obtained by bootstrapping the entire two-stage procedure, thereby capturing both sampling variation and machine-learning uncertainty.

In summary, AIPW not only recovers a credible overall average effect of refugee peers on PISA achievement, but also delivers a transparent, robust pathway to estimate how that effect differs across economically meaningful subgroups.

6 Empirical Results

The empirical discussion proceeds in four steps. We begin with a baseline difference-in-differences (DiD) specification, extend it with an instrumental-variables (IV) variant, then adopt a doubly-robust Augmented Inverse-Probability Weighting (AIPW) design to recover average treatment effects (ATE), and finally probe heterogeneity through conditional average treatment effects (CATE). Presenting the models in this sequence makes clear how each layer of identification sharpens—or tempers—the statistical story about Syrian-refugee peer exposure in Canadian PISA data.

6.1 Baseline Difference-in-Differences

Table 8 shows that DiD coefficients are negative across all three disciplines, but none approach conventional significance. Math posts the largest point estimate at -0.15 standard deviations (SD) with a t-statistic of -1.55; reading and science are -0.07 and -0.10 SD, respectively, with t-ratios comfortably below significance. Effects seem slightly more pronouced for males, but are not conclusive without further reseach due to the lack of statistical significance. Pre-treatment trend tests in Tables 5 and 6 confirm that treated and control strata followed similar trajectories before the refugee inflow, so the imprecision likely reflects sampling variability rather than design failure. Although gender-split figures 2 and 3 hint that boys may be marginally more affected than girls, the overlapping confidence bands suggest that any such differences are modest. We see little to no consistent diference in science at first glance looking at figure 4.

6.2 Instrumental-Variables Extension

Using the DiD treatment indicator as an instrument for the actual immigrant share introduces an additional safeguard against endogenous sorting, but the first-stage relationship is weak: the F-statistic falls below the conventional threshold of ten once region dummies are included. Unsurprisingly, the second-stage estimates for immigrant share in Tables 9 and 10 have such large standard errors than render the coefficient estimates meaningless, and a bootstrap Wald ratio based on 500 resamples yields similarly attenuated effects as shown in table 13. The IV exercise therefore provides no decisive evidence of either harm or benefit beyond the baseline DiD.

6.3 Average Treatment Effects via AIPW

Moving to the doubly-robust AIPW framework, Table 14 suggests a negative impact on mathematics (-0.08 SD) and positive impacts on reading ($+0.10$ SD) and science ($+0.05$ SD), all apparently significant at the one-per- cent level. These results, however, rely on naïve standard errors that ignore the complex PISA sample design. Once balanced-repeated-replication (BRR) weights are brought in (Table 15), none of the three coefficients remains significant; retaining school-clustered but not BRR standard errors (Table 16) leaves only the reading estimate significant ($+0.10$ SD). The comparison underscores the sensitivity of global ATE conclusions to the way sampling uncertainty is handled.

6.4 Conditional Average Treatment Effects

Tables 17–19 investigate whether peer effects are concentrated in particular sub-groups. Girls experience a sizeable and statistically robust penalty in mathematics (-0.19 SD) and a smaller loss in reading (-0.09 SD), while boys enjoy a significant gain in reading ($+0.13$ SD) and no statistically discernible change in mathematics. Turning to socio-economic status, the lowest ESCS quartile gains $\approx +0.10$ SD in reading, whereas the second quartile suffers a -0.15 SD drop in math; the upper half of the ESCS distribution generally shows no statistically significant material effects other than a potential decrease in math for the highest quartile. Finally, disaggregation by urbanicity produces a decreasing positive-to-negative effect on reading and a monotone downturn in mathematics: schools in smaller centres with lower populations reap gains of roughly $+0.16$ SD in reading, whereas large-city schools absorb losses of -0.14 SD in reading and -0.23 SD in math (Table 19).

Observing Figures 6, 7, and 8, the positive effects are most accentuated for students in ESCS quartile 4 and less urban settings, with an increase in reading and science which is greater for males than females. Females in ESCS quartile 4 and in the densest urban settings, however, are shown to experience negative effects in math, reading, and science. In addition, we observe statistically significant heterogeneity not only between gender but within the same gender and ESCS bucket, as seen with women who gain reading skills in moderately rural areas contrasted with the women of the same bucket losing reading skills if living in a highly urban area.

It is important to exercise caution in these results, however, remembering that unlike the global ATE analysis, the CATE exercise does not fully exploit PISA's BRR replicate weights; instead it relies on cross-fitted AIPW scores and school-clustered standard errors. Because replicate variance tends to be larger, the standard errors reported here are likely understated, meaning that the sub-group effects—while suggestive—may appear more extreme than they truly are once full design uncertainty is taken into account. Finally, through figure 9 we see a contrast between men and women based on urbanness with women's reading and science scores decreasing sharply in more urban areas whereas men's are less affected. Furthermore we see a strong increase in science scores among males in less rural areas, hinting that the facilities that refugees benefit from could have certain spillover effects in less resource rich rural areas.

6.5 Putting the Pieces Together

Taken as a whole, the results portray a nuanced landscape. On average, refugee peer exposure leaves Canadian students largely unaffected once one respects the survey design, but this overall neutrality masks marked heterogeneity. Low-income boys in semi-urban schools benefit in reading, possibly because heightened labour-market competition raises the perceived return to language skills. Another possibility is the positive spillover effects of the new investments into educating refugees disproprotionately affecting males when it comes to the disproportionate benefits they experience in rural treated areas. At the same time, girls in densely populated schools pay a clear price in mathematics, a pattern consistent with instructional crowd-out as teachers divert time toward language support. This compounded with the general stigma of women in STEM seems to catalyze as a statistically significant drop in scores. These findings caution against one-size-fits-all pronouncements due to the heterogeneity we observe among each subjection through our CATE analysis. As such, effective policy will require targeted language resources where gains are plausible and numeracy safeguards where harm is most likely, all the while bearing in mind that the extreme values reported here may soften once replicate-weight uncertainty is fully incorporated.

7 Conclusion

Syrian resettlement in Canada created a quasi-experimental setting in which two forces, the positive labour-market incentive and the negative classroom-congestion channel, act simultaneously on native students. In the aggregate the forces nearly cancel, so the mean impact on PISA scores hovers statistically near zero once we respect the survey's balanced-repeated-replication design. Because the weak-instrument two-stage least squares adds little precision, with its first-stage F barely exceeding 0.1 after covariate saturation, we lean on the doubly-robust, cross-fitted AIPW evidence when drawing policy lessons. The Augmented Inverse-Probability Weighting (AIPW) estimates reveal that neutrality is a macro illusion produced by strong but offsetting conditional effects. Reading scores rise by roughly 0.16 SD for low-SES boys attending semi-urban schools, whereas mathematics scores fall by about 0.23 SD for girls in large-city schools.

The heterogeneous pattern is straightforward to interpret within the production function sketched in imagined in section 2. Let

$$\tau(x) = \tau_{\text{labour}}(x) - \tau_{\text{congestion}}(x)$$

denote the net peer effect for a student whose background vector is x. Where labour-market competition is salient for boys whose outside option is immediate entry into low-skill employment we observe $\tau_{\text{labour}}(x)$ to be the dominating effect. Furthermore, where instructional resources are scarce—urban classrooms that suddenly accommodate newcomers we observe $\tau_{\text{congestion}}(x)$ is a dominating negative effect. Our CATE estimates show that the first term dominates for disadvantaged boys in smaller centres, while the second term dominates for girls in metropolitan schools. Theoretically, then, equity-enhancing policy must either elevate τ_{labour} for girls, depress $\tau_{\text{congestion}}$ where it bites hardest, or accomplish both.

Several mutually reinforcing strategies emerge from these observations. Provincial ministries could issue temporary, refugee-adjusted class-size caps in mathematics, fund team-teaching or split-grade arrangements, and underwrite the hiring of Arabic-speaking instructional aides who allow mainstream teachers to maintain lesson pace without sacrificing newcomer support. Where staffing budgets make permanent aides unrealistic, the congestion term can be dampened through low-cost technologies: digital platforms that let refugee pupils preview or review mathematics content in Arabic or Kurmanji, and AI-assisted captioning that supplies real-time glossaries of technical terms. By returning instructional minutes to the core curriculum, these measures push $\tau_{\text{congestion}}(x)$ toward zero without muting the competitive spur captured by $\tau_{\text{labour}}(x)$.

Yet congestion relief alone is insufficient if girls never perceive the labour-market premium that motivates their male classmates. Small, precisely timed nudges can raise the incentive component for female students. Embedding a one-page wage-return vignette in ninth-grade report cards, hosting lunchtime webinars with local women working in data science or engineering, and guaranteeing transportation stipends for after-school STEM clubs all lower the information and access barriers that keep $\tau_{\text{labour}}(x)$ artificially low for girls. Because these interventions operate on expectations rather than resources, they are cheap to scale and complementary to the structural fixes above.

The policy recipe that emerges is therefore two-dimensional: preserve or even amplify the competitive channel that appears salutary for disadvantaged boys, while selectively neutralising the classroom-congestion channel that harms urban girls in numeracy-intensive subjects. Algebraically, the goal is to make

$$\tau_{\text{labour}}(x) \approx \tau_{\text{congestion}}(x) \quad \text{for all } x,$$

so that the conditional average treatment effect converges to a benign mean of zero without masking pockets of loss.

At the same time, we must observe three cautions to these conclusions. First, the neutrality we observe is contingent on Canada's modest intake rate; larger or more geographically concentrated arrivals could tilt the balance decisively negative unless compensatory resources scaled in lock-step. Secondly, Canada has a highly selected group of refugees compared to those in countries neighboring a conflict, who absorb waves of people so fast that there is no time for proper processing and cracks in the wall emerge. Thirdly, the present analysis is cross-sectional. Whether the short-run reading gains and mathematics losses persist into high-school completion, post-secondary choice, and labour-market entry remains unknown. Linking future PISA cohorts to longitudinal administrative data would permit a full difference-in-differences-in-cohorts design that traces refugee spill-overs beyond age fifteen.

Even so, the central lesson is robust: humanitarian admissions need not erode host-country human capital, but the distribution of costs and benefits is sharply uneven. If policy makers couple agile resource allocation with gender-sensitive incentive nudges, they can turn potential congestion into a labouratory of civic learning, ensuring that the next wave of refugee children enlarges rather than diminishes the opportunities of their Canadian peers.

Finally, a natural extension of this paper is to wonder about 2022's data and its use in other countries which experience only the peer effect such as Austria and Denmark. Both are countries with incredibly strict laws with working under 18 and so the labour effect is significantly dampened in these stats. As such, we can compare Canada with these two in an attempt to try to isolate both effects and see their interplay more so than we already do. In addition, the extra year of PISA data would help lower our standard errors and get more robust final BRR results.

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8 Appendix

Table 4: Parallel Trends Regression Results (Standardized Coefficients)

Discipline	Group	Coefficient	Std. Error
Math	Control	-0.196	$+0.061$
	Treatment	-0.075	$+0.049$
Reading	Control	-0.217	$+0.065$
	Treatment	$+0.060$	$+0.048$
Science	Control	-0.179	$+0.063$
	Treatment	$+0.024$	$+0.045$

Note: Standardized results of our parallel trend analysis without stratum level fixed effects between 2012 and 2015 across disciplines.

Table 5: Parallel Trends Regression Results (Standardized Coefficients)

Discipline	Group	Coefficient	Std. Error
Math	Control	-0.045	$+0.069$
	Treatment	$+0.012$	$+0.047$
Reading	Control	$+0.037$	$+0.066$
	Treatment	$+0.066$	$+0.050$
Science	Control	-0.022	$+0.058$
	Treatment	$+0.073$	$+0.044$

Note: Standardized results of our parallel trend analysis with stratum level fixed effects between 2012 and 2015 across disciplines.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Placebo Regression Results (Standardized Coefficients, 2012–2015 Only)

Discipline	Coefficient	Std. Error	t-statistic
Math	+0.044	0.075	+0.57
Reading	+0.017	0.075	+0.22
Science	+0.072	0.070	1.03

Note: Standardized results of placebo effect regression using 2012 and 2015 PISA data.

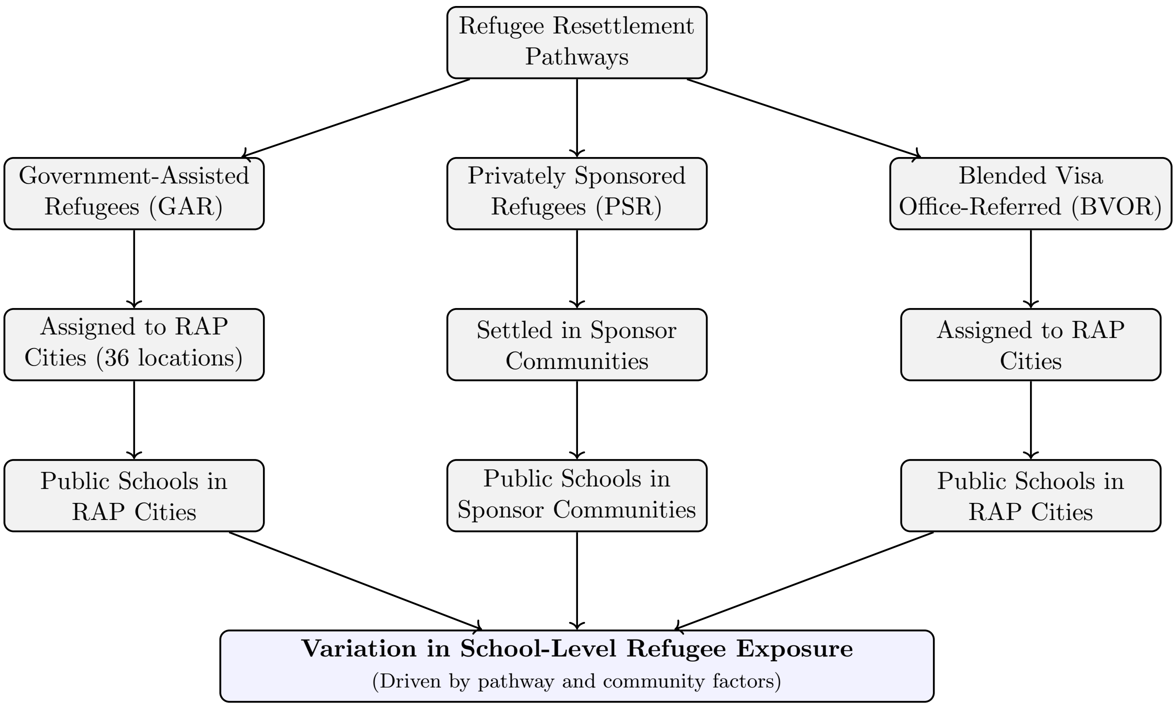


Figure 1: Structure of Canada's Refugee Resettlement Pathways and Their Impact on School-Level Refugee Exposure

Table 7: Predicting 2018 Refugee Assignment Using 2015 Stratum-Level Characteristics

Variable	**Coefficient**	**Std. Error**	**t-statistic**
% Stratum Urbanity	1.34***	0.25	5.30
% First-Gen Immigrants	3.20***	0.93	3.45

Note: Both the % Urbanity and % First-Gen Immigrants are on the stratum level. This regression serves to show our ability to predict if the stratum will be treated in the next period given the stratum's current urbanization and density of first generation immigrants.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Difference-in-Differences Results (Standardized Estimates)

Discipline	**Group**	**Coefficient (SD)**	**Std. Error (SD)**	**t-statistic**
Math	All	−0.150	0.097	−1.552
	Female	−0.134	0.110	−1.224
	Male	−0.185	0.107	−1.726
Reading	All	−0.072	0.094	−0.769
	Female	−0.067	0.103	−0.654
	Male	−0.105	0.101	−1.042
Science	All	−0.095	0.087	−1.090
	Female	−0.108	0.099	−1.083
	Male	−0.121	0.098	−1.223

Note: Table presents the estimated effects of refugee peer exposure on standardized PISA scores across three academic disciplines using a difference-in-differences (DiD) framework. The regressions include school, stratum, and survey-year fixed effects to control for unobserved heterogeneity across schools, regional assignment structures, and national trends, respectively. Additional controls include students' wealth status and maternal education.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

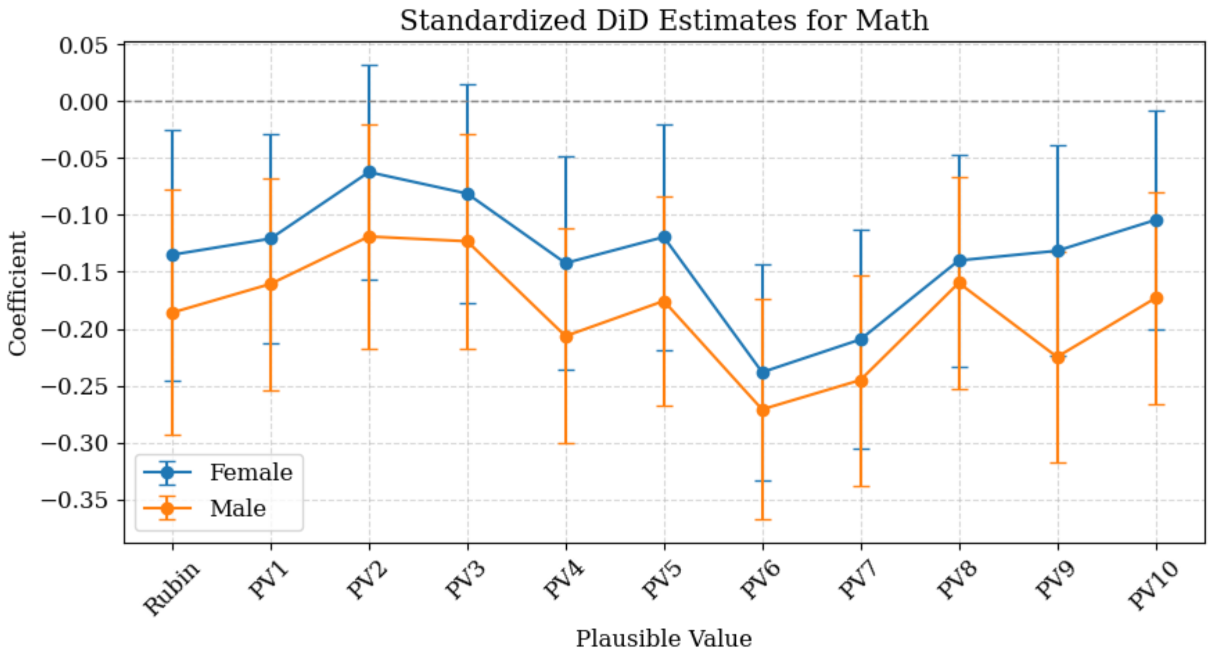


Figure 2: Standardized DiD Results for Math by Plausible Value and Gender

Note: Coefficient results represent treatment effects for math in theoretical PISA standard deviation units, disaggregated by plausible value (PV1–PV10, Rubin Combined) and by gender (male and female). Tails represent standard errors.

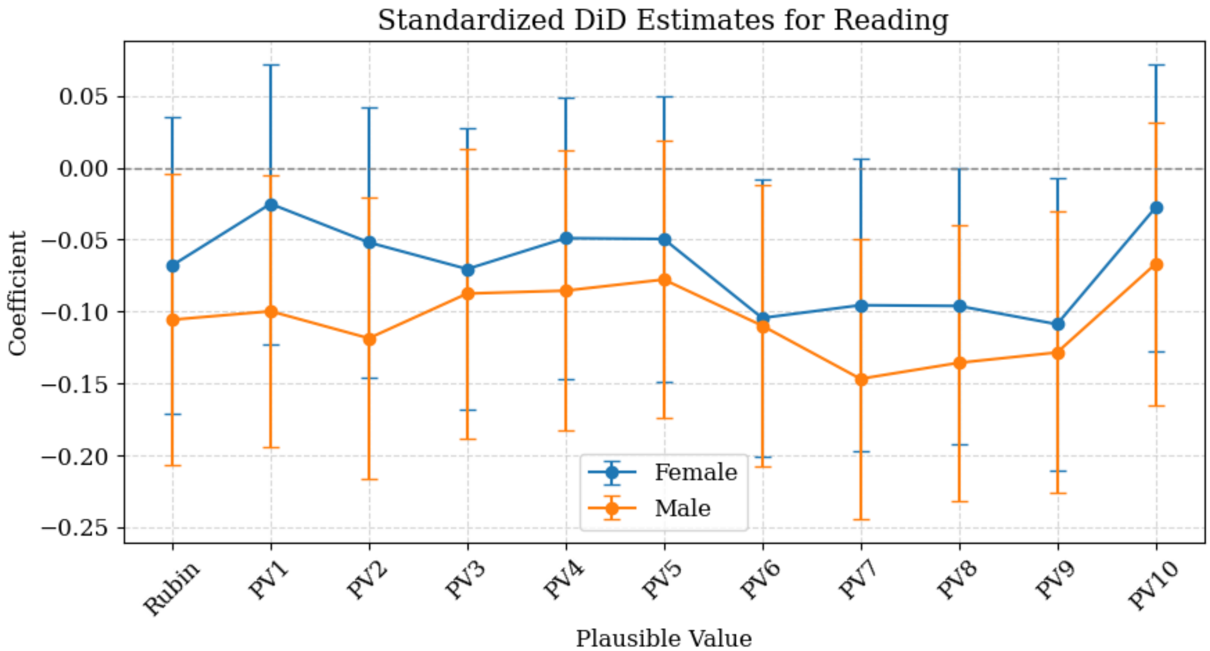


Figure 3: Standardized DiD Results for Reading by Plausible Value and Gender
Note: Coefficient results represent treatment effects for reading in theoretical PISA standard deviation units, disaggregated by plausible value (PV1–PV10, Rubin Combined) and by gender (male and female). Tails represent standard errors.

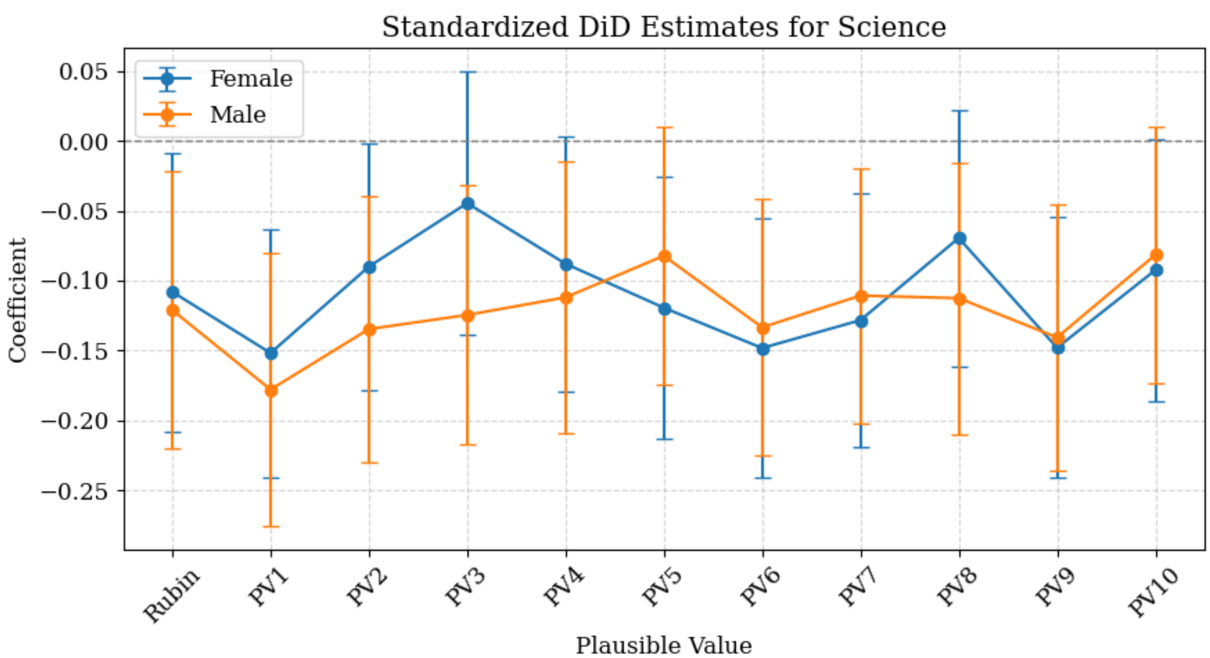


Figure 4: Standardized DiD Results for Science by Plausible Value and Gender
Note: Coefficient results represent treatment effects for science in theoretical PISA standard deviation units, disaggregated by plausible value (PV1–PV10, Rubin Combined) and by gender (male and female). Tails represent standard errors.

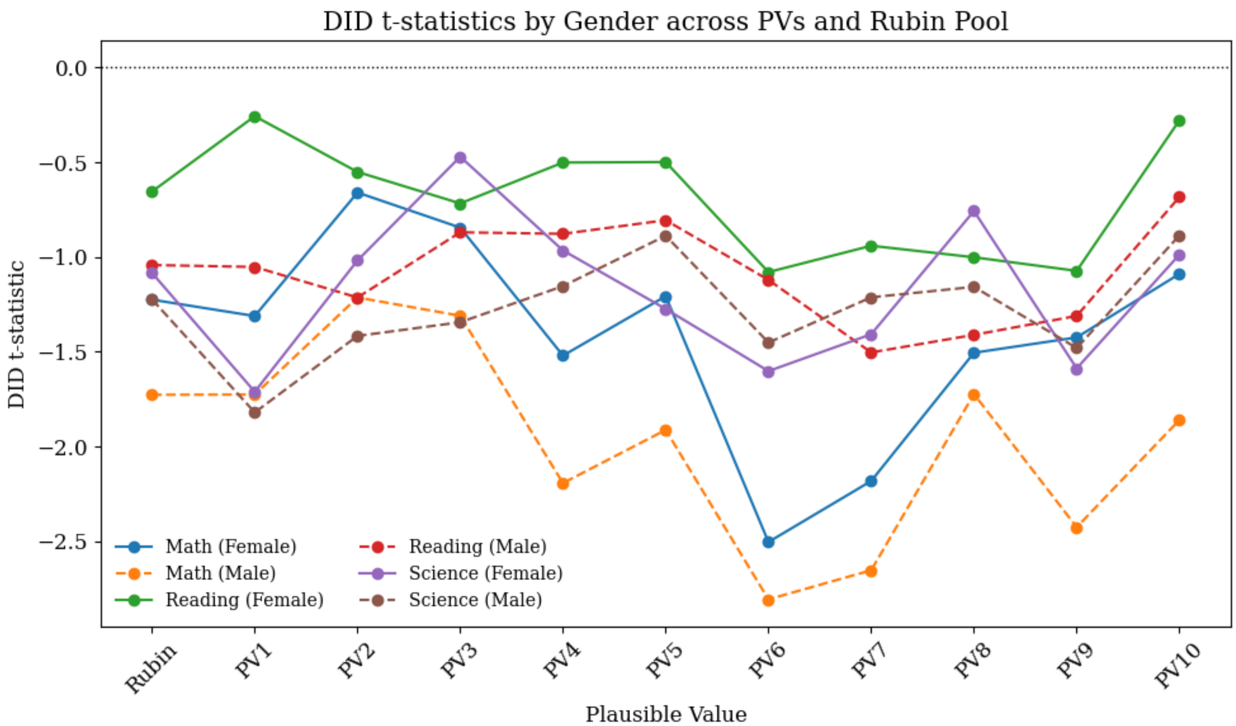


Figure 5: t-stat results for DiD by PV and gender

Note: T-statistics of standardized difference in difference model across plausible values by gender and subject

Table 9: PV1: 2SLS with Covariates and No Weights

Variable	First Stage	Reduced form Math	Reduced form Read	Reduced form Science	Second stage Math	Second stage Read	Second stage Science
Intercept	0.011***	-0.976***	-0.897***	-0.834***	-0.954***	-1.003***	-0.852***
	(0.003)	(0.037)	(0.041)	(0.039)	(0.046)	(0.066)	(0.049)
TREATMENT_SPACE	0.047***	0.169***	0.234***	0.191***	0.261	-0.203	0.117
	(0.002)	(0.027)	(0.031)	(0.029)	(0.176)	(0.265)	(0.189)
TREATMENT_TIME	0.013***	-0.007	-0.145***	-0.098**	0.019	-0.269**	-0.119
	(0.002)	(0.026)	(0.037)	(0.030)	(0.066)	(0.107)	(0.074)
DID	0.011***	-0.021	0.100**	0.017			
	(0.003)	(0.036)	(0.044)	(0.038)			
MISCED	-0.001***	0.135***	0.140***	0.140***	0.133***	0.150***	0.142***
	(0.000)	(0.005)	(0.005)	(0.005)	(0.006)	(0.008)	(0.006)
WEALTH	0.002***	0.035***	0.017**	0.019**	0.039***	-0.002	0.016
	(0.000)	(0.006)	(0.006)	(0.006)	(0.009)	(0.012)	(0.009)
URBAN	0.034***	0.348***	0.299***	0.265***	0.415**	-0.019	0.211
	(0.004)	(0.041)	(0.044)	(0.042)	(0.122)	(0.180)	(0.127)
IMMIGRANT_SHARE					-1.963	9.267	1.569
					(3.333)	(5.048)	(3.555)
r2	0.452	0.054	0.057	0.048	0.056	-0.055	0.042
N	37644	37644	37644	37644	37644	37644	37644

Note: 2SLS with school clustered standard errors on PV1 using no weights or PISA methodology
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: PV1: 2SLS with PISA Methodology

	First Stage	Reduced form			Second stage		
Variable		Math	Read	Science	Math	Read	Science
Intercept	0.021***	-0.904***	-0.810***	-0.737***	-0.836	-1.484	-0.887
	(0.006)	(0.088)	(0.087)	(0.092)	(1.473)	(8.626)	(1.762)
TREATMENT_SPACE	0.049***	0.192**	0.163*	0.141*	0.353	-1.424	-0.214
	(0.005)	(0.060)	(0.064)	(0.071)	(3.445)	(20.515)	(4.296)
TREATMENT_TIME	0.023**	-0.044	-0.154*	-0.141	0.033	-0.912	-0.310
	(0.009)	(0.066)	(0.077)	(0.075)	(1.766)	(10.391)	(2.132)
DID	0.002	-0.008	0.076	0.017			
	(0.009)	(0.077)	(0.087)	(0.087)			
MISCED	-0.000	0.140***	0.144***	0.144***	0.139***	0.159	0.148**
	(0.000)	(0.011)	(0.010)	(0.010)	(0.033)	(0.212)	(0.046)
WEALTH	0.002***	0.016	0.008	0.008	0.023	-0.063	-0.008
	(0.000)	(0.011)	(0.013)	(0.012)	(0.158)	(0.922)	(0.188)
URBAN	0.021***	0.286**	0.340***	0.269**	0.357	-0.357	0.113
	(0.006)	(0.087)	(0.089)	(0.086)	(1.401)	(8.399)	(1.679)
IMMIGRANT_SHARE					-3.289	32.463	7.263
					(69.679)	(412.373)	(84.834)
r2	0.307	0.044	0.043	0.041	0.044	0.043	0.041
N	37675	37675	37675	37675	37675	37675	37675

Note: 2SLS with school clustered standard errors on PV1 using PISA methodology
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 11: PV1–PV10: 2SLS MI-pooled results with covariates and cluster-robust SEs

	First Stage	Reduced form			Second stage		
Variable		Math	Read	Science	Math	Read	Science
Intercept	0.011***	-0.981***	-0.911***	-0.829***	-0.982***	-1.026***	-0.874***
	(0.003)	(0.050)	(0.049)	(0.045)	(0.068)	(0.074)	(0.060)
TREATMENT_SPACE	0.047***	0.132***	0.235***	0.204***	0.125	-0.240	0.018
	(0.002)	(0.037)	(0.035)	(0.032)	(0.236)	(0.297)	(0.230)
TREATMENT_TIME	0.013***	-0.020	-0.153***	-0.129***	-0.022	-0.288*	-0.181
	(0.002)	(0.040)	(0.046)	(0.039)	(0.096)	(0.124)	(0.096)
DID	0.011***	0.002	0.109*	0.042			
	(0.003)	(0.049)	(0.049)	(0.045)			
MISCED	-0.001***	0.139***	0.141***	0.137***	0.139***	0.151***	0.141***
	(0.000)	(0.007)	(0.007)	(0.006)	(0.008)	(0.009)	(0.007)
WEALTH	0.002***	0.036***	0.016	0.022**	0.035**	-0.005	0.014
	(0.000)	(0.007)	(0.008)	(0.007)	(0.012)	(0.015)	(0.011)
URBAN	0.034***	0.364***	0.304***	0.266***	0.359*	-0.042	0.130
	(0.004)	(0.049)	(0.047)	(0.048)	(0.159)	(0.201)	(0.158)
IMMIGRANT_SHARE					0.144	10.072	3.933
					(4.564)	(5.662)	(4.431)
r2	0.452	0.054	0.058	0.050	0.045	-0.075	0.024
N	37644	37644	37644	37644	37644	37644	37644

Note: 2SLS with school clustered standard errors on all plausible values using no weights and Rubin's rules.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 12: PV1–PV10: 2SLS MI-pooled results with BRR–Fay and cluster-robust SEs

Variable	First Stage	Reduced form Math	Reduced form Read	Reduced form Science	Second stage Math	Second stage Read	Second stage Science
Intercept	0.021***	-0.921***	-0.843***	-0.744***	-0.624	-1.385	-1.060
	(0.006)	(0.101)	(0.096)	(0.095)	(5.467)	(7.033)	(4.836)
TREATMENT_SPACE	0.049***	0.178**	0.192**	0.163*	0.877	-1.085	-0.579
	(0.005)	(0.069)	(0.070)	(0.072)	(12.957)	(16.604)	(11.613)
TREATMENT_TIME	0.023**	-0.018	-0.140	-0.154	0.317	-0.749	-0.508
	(0.009)	(0.081)	(0.080)	(0.080)	(6.611)	(8.433)	(5.845)
DID	0.002	-0.034	0.061	0.036			
	(0.009)	(0.089)	(0.091)	(0.092)			
MISCED	-0.000	0.142***	0.145***	0.143***	0.136	0.158	0.151
	(0.000)	(0.012)	(0.011)	(0.011)	(0.131)	(0.171)	(0.122)
WEALTH	0.002***	0.014	0.006	0.009	0.046	-0.051	-0.024
	(0.000)	(0.013)	(0.014)	(0.012)	(0.588)	(0.749)	(0.517)
URBAN	0.021***	0.307**	0.338***	0.257**	0.614	-0.223	-0.069
	(0.006)	(0.098)	(0.095)	(0.089)	(5.302)	(6.773)	(4.755)
IMMIGRANT_SHARE					-14.304	26.107	15.188
					(261.016)	(334.261)	(232.566)
r2	0.307	0.044	0.045	0.042	0.044	0.045	0.042
N	37675	37675	37675	37675	37675	37675	37675

Note: 2SLS with school clustered and BRR standard errors on all plausible values using full PISA methodology.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 13: Wald Estimator Results by Discipline and Plausible Value (Standardized)

PV	**Math**			**Reading**			**Science**		
	Wald β	SE	p	Wald β	SE	p	Wald β	SE	p
Rubin	-0.093	0.226	0.681	-0.011	0.230	0.961	-0.032	0.228	0.888
PV1	-0.093	0.257	0.719	0.007	0.255	0.977	-0.074	0.258	0.774
PV2	-0.037	0.210	0.859	-0.012	0.243	0.962	-0.027	0.236	0.908
PV3	-0.030	0.204	0.884	-0.023	0.239	0.922	-0.007	0.229	0.976
PV4	-0.075	0.246	0.759	0.021	0.250	0.932	-0.020	0.242	0.934
PV5	-0.100	0.230	0.663	0.030	0.237	0.898	-0.004	0.237	0.986
PV6	-0.182	0.300	0.544	-0.032	0.250	0.898	-0.070	0.247	0.776
PV7	-0.130	0.295	0.660	-0.040	0.245	0.870	-0.035	0.239	0.885
PV8	-0.102	0.260	0.694	-0.033	0.252	0.895	0.002	0.239	0.993
PV9	-0.117	0.270	0.664	-0.053	0.260	0.840	-0.066	0.281	0.813
PV10	-0.064	0.225	0.775	0.020	0.241	0.935	-0.020	0.244	0.934

Note: Coefficients and standard errors are scaled by 1/100. Standard errors are calculated using Balanced Repeated Replication (BRR) clustered at the school level. Wald estimates reflect the effect of a one percentage point increase in first-generation immigrant share—attributable to refugee resettlement—on host students' standardized PISA test performance.

Table 14: AIPW Estimates of Refugee Impact on PISA Scores (ATE)

Discipline	**ATE**	**Std. Error**	**t-statistic**	**p-value**
Math	-0.081	0.018	-4.617	< 0.001
Reading	0.096	0.006	16.735	< 0.001
Science	0.051	0.013	3.796	< 0.001

Note: Estimates are obtained using Augmented Inverse Probability Weighting (AIPW) with standard PISA final student weights only. No BRR replicate weights or clustered standard errors are applied. Results represent the estimated causal effect of refugee presence on host students' standardized PISA scores by discipline.

Table 15: AIPW Estimates of Refugee Impact on PISA Scores (ATE)

Discipline	**ATE**	**Std. Error**	**t-statistic**	**p-value**
Math	-0.092	0.120	-0.77	0.441
Reading	0.073	0.085	0.859	0.390
Science	0.041	0.097	0.424	0.671

Note: Estimates are obtained using Augmented Inverse Probability Weighting (AIPW) with the full PISA methodology using BRR replicate weights. Results represent the estimated causal effect of refugee presence on host students' standardized PISA scores by discipline.

Table 16: AIPW Estimates of Refugee Impact on PISA Scores (ATE)

Discipline	**ATE**	**Std. Error**	**t-statistic**	**p-value**
Math	-0.063	0.078	-0.82	0.414
Reading	0.095	0.034	2.75	0.006
Science	0.067	0.049	1.37	0.171

Note: Estimates are obtained using Augmented Inverse Probability Weighting (AIPW) with school clustered standard errors. Results represent the estimated causal effect of refugee presence on host students' standardized PISA scores by discipline.

Table 17: Conditional Average Treatment Effects (CATE) by Gender and Discipline

Gender	**Discipline**	**CATE**	**Std. Error**	**t-statistic**	**p-value**
Female (ST004D01T = 1)	Math	−0.190	0.051	−3.71	0.000
	Reading	−0.094	0.037	−2.52	0.012
	Science	−0.085	0.054	−1.56	0.119
Male (ST004D01T = 2)	Math	−0.059	0.074	−0.80	0.426
	Reading	0.131	0.033	3.98	0.000
	Science	0.069	0.044	1.56	0.119

Note: Estimated treatment effects are separated by gender (male, female) and academic discipline (e.g., math, reading, science).

Table 18: Conditional Average Treatment Effects (CATE) by ESCS Quartile and Discipline

ESCS Quartile	**Discipline**	**CATE**	**Std. Error**	**t-statistic**	**p-value**
Q1 (Lowest)	Math	−0.099	0.087	−1.14	0.256
	Reading	0.101	0.050	2.00	0.045
	Science	0.024	0.074	0.33	0.744
Q2	Math	−0.150	0.072	−2.07	0.038
	Reading	−0.044	0.047	−0.93	0.352
	Science	−0.045	0.073	−0.62	0.532
Q3	Math	−0.132	0.084	−1.57	0.116
	Reading	0.019	0.053	0.36	0.720
	Science	−0.003	0.057	−0.05	0.957
Q4 (Highest)	Math	−0.100	0.060	−1.66	0.097
	Reading	0.004	0.053	0.07	0.942
	Science	0.013	0.063	0.21	0.835

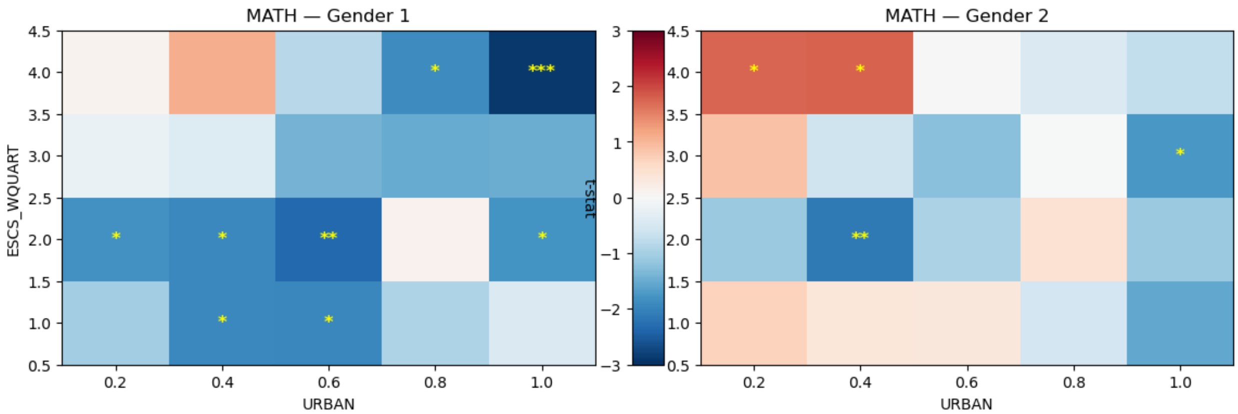
Note: Estimated treatment effects are separated by ESCS quartiles (with increasing quartiles representing increasing ESCS status) and academic discipline (e.g., math, reading, science).

Table 19: Conditional Average Treatment Effects (CATE) by Urban Level and Domain

Urban	**Discipline**	**CATE**	**SE_clustered**	**t-stat**	**p-value**
	MATH	-0.085946	0.139857	-0.614525	0.538869
0.2	READING	0.172180	0.070989	2.425463	0.015289
	SCIENCE	0.045698	0.087849	0.520186	0.602934
	MATH	-0.122478	0.126188	-0.970599	0.331748
0.4	READING	0.158173	0.058870	2.686831	0.007213
	SCIENCE	0.029894	0.064655	0.462367	0.643818
	MATH	-0.141025	0.100675	-1.400793	0.161276
0.6	READING	0.065701	0.052019	1.263001	0.206589
	SCIENCE	-0.021375	0.069518	-0.307480	0.758478
	MATH	-0.067459	0.057915	-1.164800	0.244100
0.8	READING	-0.008685	0.048520	-0.179001	0.857937
	SCIENCE	0.023952	0.054981	0.435648	0.663092
	MATH	-0.226478	0.094884	-2.386895	0.016991
1.0	READING	-0.143104	0.071064	-2.013733	0.044038
	SCIENCE	-0.105990	0.090820	-1.167029	0.243199

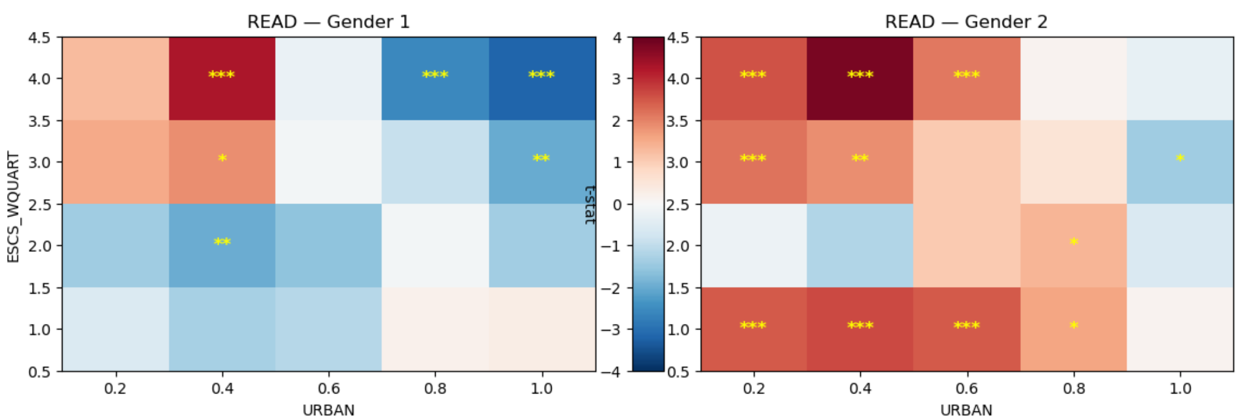
Note: Estimated treatment effects are separated by urban level (with increasing urban scoring representing a greater population density) and academic discipline (e.g., math, reading, science).

Figure 6: Heatmap of CATE t-statistics by urbanneess and ESCS quartile separate by gender (Math)



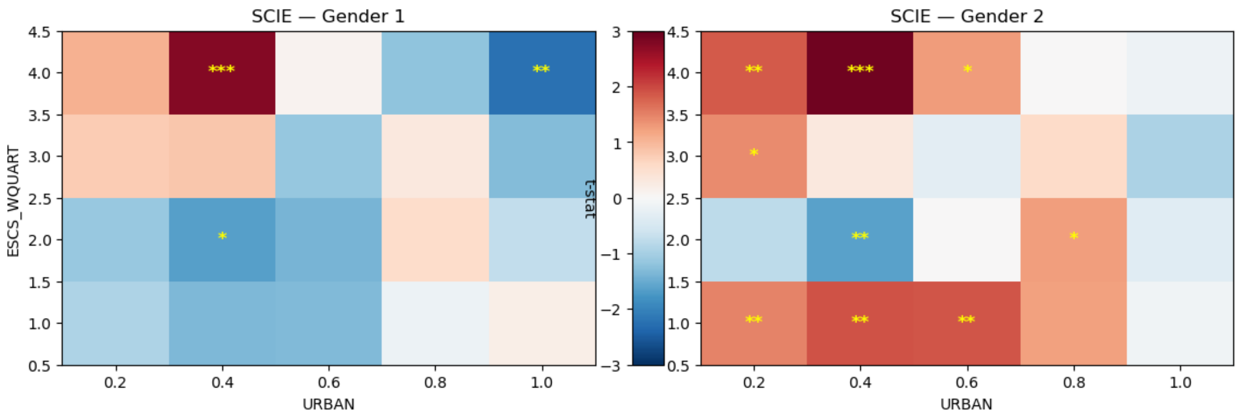
Note: T-statistics for math (represented by blue hues for positive and red hues for negative) are separated by gender and organized by ESCS quartiles (a discrete value Q1 to Q4, by increasing ESCS status) and urban level (with increasing urban scoring representing a greater population density).
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 7: Heatmap of CATE t-statistics by urbanneess and escs quartile separate by gender (Reading)



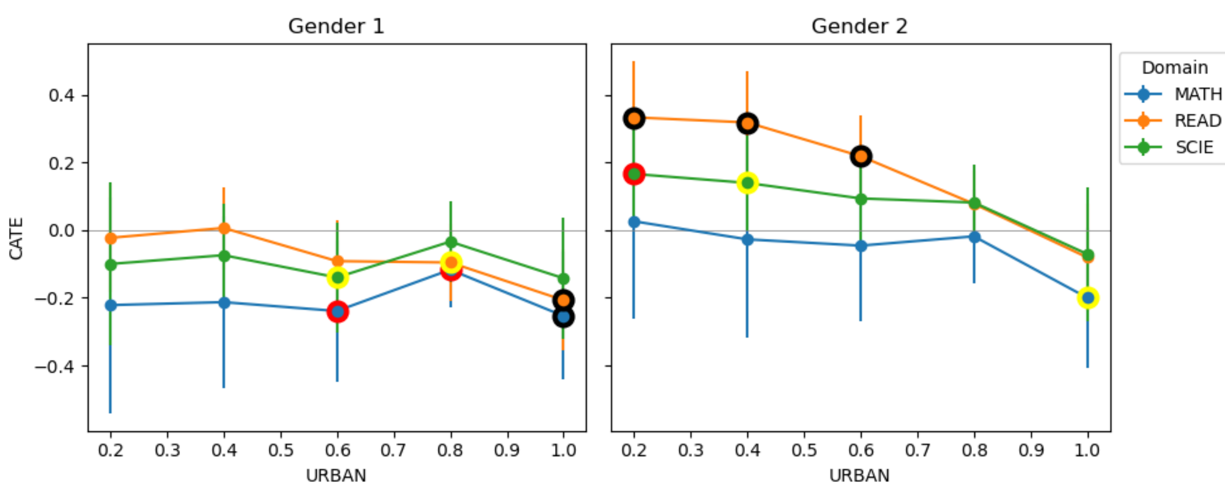
Note: T-statistics for reading (represented by blue hues for positive and red hues for negative) are separated by gender and organized by ESCS quartiles (a discrete value Q1 to Q4, by increasing ESCS status) and urban level (with increasing urban scoring representing a greater population density).
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 8: Heatmap of CATE t-statistics by urbanneess and escs quartile separate by gender (Science)



Note: T-statistics for science (represented by blue hues for positive and red hues for negative) are separated by gender and organized by ESCS quartiles (a discrete value Q1 to Q4, by increasing ESCS status) and urban level (with increasing urban scoring representing a greater population density).
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 9: Conditional Average Treatment Effect of School Level Refugee Presence by Urbanness and Gender



Note: Average treatment effects for different genders (Gender 1 = Females, Gender 2 = Males) are separated by discipline and urban levels (with increasing urban scoring representing a greater population density). Circles around point estimates indicate significance: Yellow — $p < 0.1$, Red — $p < 0.05$, Black — $p < 0.01$